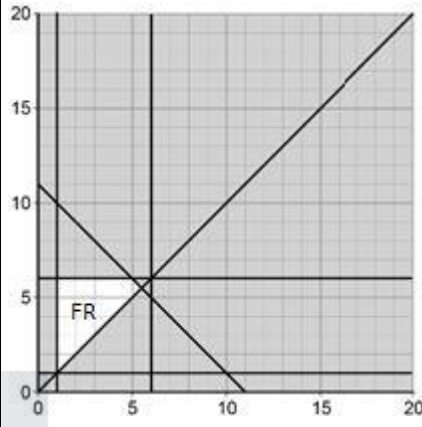


Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Draws $y = x$	AO1.1a	M1	
	Clearly labels (or shows by shading) the correct feasible region	AO1.1b	A1	
(ii)	Calculates the correct value of $5x + 4y$ at $(5, 6)$ or $(5.5, 5.5)$	AO1.1a	M1	Value at $(5, 6) = 49$ Value at $(5.5, 5.5) = 49.5$
	Finds the correct maximum value of $5x + 4y$	AO1.1b	A1	Maximum value is 49.5
(b) (i)	Explains the implication of the graph being connected Must refer explicitly to 'connected'	AO1.2	B1	The graph is connected, so no vertex can have degree 0.
	Explains the implication of the graph being simple Must refer explicitly to 'simple'	AO1.2	B1	The graph is simple, so each vertex can only connect at most once to each of the 6 other vertices.
(iii)	Explains why x and y cannot both be 6	AO2.4	E1	If x and y are both 6 then there would be two vertices that are both adjacent to the other five vertices.
	Completes rigorous argument for $x + y \leq 11$	AO2.1	R1	However, this would mean that there would be no vertex of degree 1 so one of x or y has to be at most 5. Hence $x + y \leq 5 + 6$
(iv)	Identifies the need for integer solutions	AO3.5b	B1	x and y must both be integers

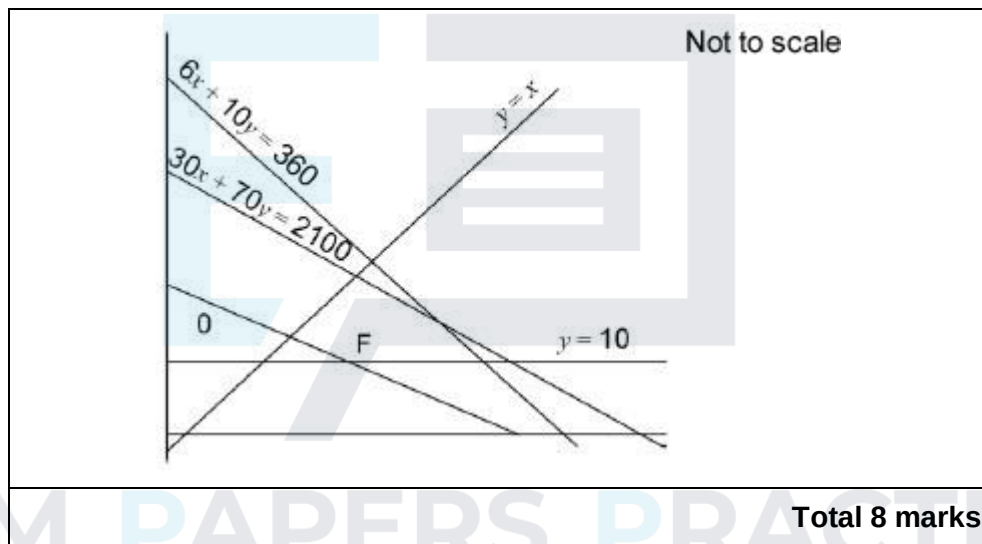
	Marking Instructions	AO	Marks	Typical Solution
(c) (i)	Uses degree sum $= 2 \times 8$ to deduce $x + y$	AO2.2a	M1	$1 + 2 + 3 + v + w + x + y = 2 \times 8$ $16 - (1 + 2 + 3 + 4) = x + y$ $x + y = 6$

	Infers there are multiple solutions and finds two correct feasible pairs of x and y values	AO2.2b	M1	$x = 1, y = 5;$ $x = 2, y = 4;$ $x = 3, y = 3.$
	Finds all three pairs and no others Condone pairs written as coordinates	AO1.1b	A1	
(ii)	Starts to show configuration of G by drawing a connected graph with exactly 8 edges OR by listing/labelling the vertex degrees: 1, 2, 2, 2, 2, 3, 4	AO3.1	M1	
	Draws a correct graph with degrees: 1, 2, 2, 2, 2, 3, 4 NB Multiple correct solutions, check by counting degrees	AO3.2a	A1	
Total 14 marks				

Q2.

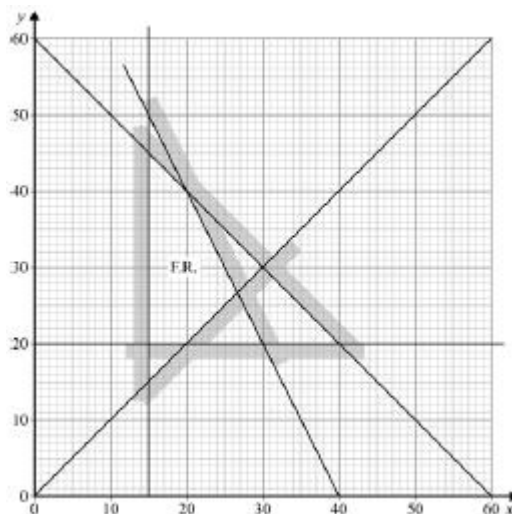
Marking Instructions	AO	Marks	Typical Solution
Introduces two variables and defines at least one of them as 'number of'	AO3.3	B1	x = number of pine tables y = number of oak tables
Finds correctly an inequality for the problem by considering the costs (PI)	AO3.1b	B1	$30x + 70y \leq 2100$ OE
Finds correctly an inequality for the problem by considering the labour (PI)	AO1.1b	B1	$6x + 10y \leq 360$ OE
Finds correctly three inequalities for the problem by considering the wholesaler (PI)	AO1.1b	B1	$y \leq x, y \geq 10, x \geq 0$
Correctly plots the line $6x + 10y = 360$ OE through (0, 36) and (60, 0)	AO1.1b	B1	*See diagram below

AND the line $30x + 70y = 2100$ OE through (0, 30) and (70, 0)			
Correctly and accurately plots 'their' $y = x$ AND 'their' $y = 10$	AO1.1b	B1	*See diagram below
Identifies and labels 'their' feasible region	AO2.2a	B1	*See diagram below
Uses the optimal vertex of the feasible region and clearly states the solution to the problem in the context of the question	AO3.2a	A1 CAO	Uses an objective line with a gradient of $-8/15$ 35 pine tables and 15 oak tables (Profit = £2525)



Q3.

(a)



Accuracy: All lines must be ruled, correct to within $\frac{1}{2}$ small square both horizontally and vertically

$$x = 15, y = 20$$

B1

$$x + y = 60,$$

correct at (10, 50) and (40, 20)

B1

$$2x + y = 80,$$

correct at (15, 50) and (30, 20)

B1

$$y = x,$$

correct at (10, 10) and (30, 30)

B1

F.R. (a pentagon) labelled, must have scored previous 4 marks

B1

5

(b) (i) (Max at) (15,45)

B1

$$(P =) 195$$

B1

2

(ii) Sight of (26 – 27, 26 – 27)

B1

$$(P =) 130 - 135$$

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M1

$$(P =) \frac{400}{3}$$

oe

A1

3

[10]

Q4.

(a) $6x + 4y + 3z \geq 420$

B1

$$6x + 6y + 4z \geq 480 \quad \text{oe}$$

B1

$$6x + 4y + 4z \leq 720$$

B1

3

(b) (i) ($y = z$)

$$6x + 4y + 3z \geq 420 \Rightarrow 6x + 7y \geq 420$$

Must see this substitution

B1

$$6x + 10y \geq 480 \Rightarrow 3x + 5y \geq 240 \quad \text{oe}$$

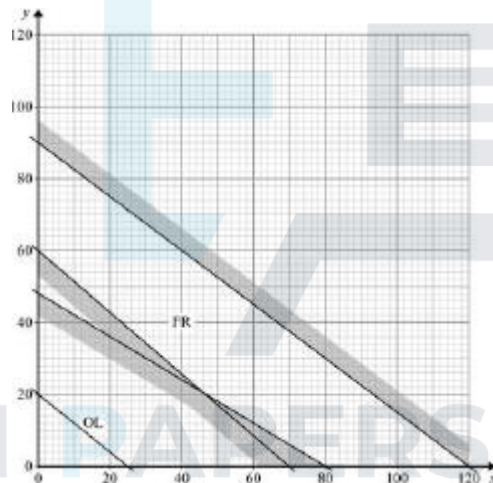
$$6x + 8y \geq 720 \Rightarrow 3x + 4y \geq 360 \quad \text{oe}$$

Both other inequalities correct, condone direct substitution into simplified versions of part (a)

B1

2

(ii)



Accuracy: All lines must be ruled, correct to within $\frac{1}{2}$ square
 BOTH horizontally and vertically
 Correct at (0, 60) and (70, 0)

B1

Correct at (0, 48) and (80, 0)

B1

Correct at (0, 90) and (120, 0)

B1

FR labelled, MUST have scored previous 3 marks

B1

Condone omission of shading on axes
OL, drawn, with gradient -0.8 or -1.25

M1

Gradient -0.8

A1

6

- (iii) (Max profit =) £480
Including '£'

B1

120 gold, 0 silver, 0 bronze
All 3 must be stated

B1

2

- (c) (Max profit =) £1080
Including '£'

B1

0 gold, 90 silver, 90 bronze
If B0 scored then B1 for $x = 0$ and $y = 90$, PI

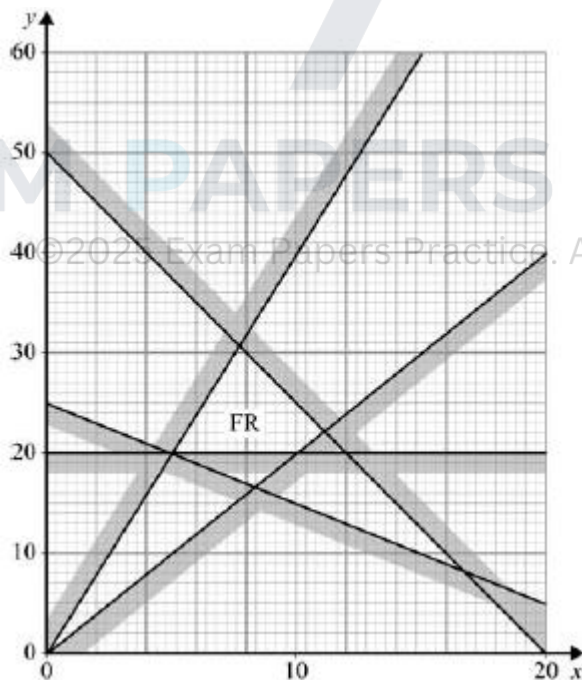
B2

3

[16]

Q5.

(a)



Each line must be straight to have the B mark available. For all lines, must be correct to $\frac{1}{2}$ square horizontal and vertical at the indicated vertices.

$$y = 20$$

B1

line through (4, 40) and (16, 10)

B1

line through (0, 25) and (10, 15)

B1

any line through origin (or if extended, through the origin)
with positive gradient (generous ± 1 square at the origin)

M1

lines through (10, 20) **and** (10, 40) as well as origin
(normal accuracy rules)

A1

FR, all lines correct and region labelled
(condone no shading, ignore 'poor' shading)

B1

6

- (b) (i) (Min at) $x = 5, y = 20$
Accept (5, 20)

B1

(P =) 45

B1

- (ii) (Min at) $x = 10, y = 20$
Accept (10, 20)

B1

(P =) 10

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B1

4

[10]

Q6.

(a)

$$\left. \begin{array}{l} x \geq 100, y \geq 200 \\ x + y + z \geq 400 \end{array} \right\} \text{OE}$$

B1

$$4x + 3y + 4z \leq 1800 \quad \text{OE}$$

B1

$$y \geq \frac{40}{100} (x + y + z) \quad \text{OE}$$

B1

3

(b) (i) $(x = 2z)$

$$\left. \begin{aligned} x + y + \frac{1}{2}x &\geq 400 \\ \Rightarrow 3x + 2y &\geq 800 \end{aligned} \right\}$$

Correct substitution and fully simplifying 1 inequality (must see evidence: either replacing z or multiplying inequality)

M1

$$\left. \begin{aligned} 4x + 3y + 2x &\leq 1800 \\ 6x + 3y &\leq 1800 \\ 2x + y &\leq 600 \end{aligned} \right\}$$

As above 'in 2nd inequality'

A1

$$\left. \begin{aligned} 5y &\geq 2x + 2y + x & 5y &\geq 2x + 2y + x \\ 3y &\geq 3x & 3y &\geq 3x \\ y &\geq x & y &\geq x \end{aligned} \right\}$$

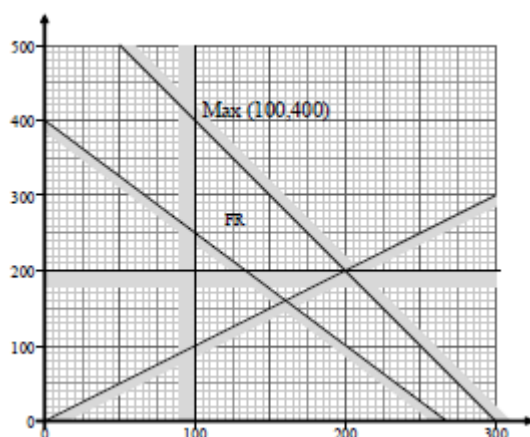
As above 'in 3rd inequality'

A1

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3

(ii)



*Each line must be straight to have the B mark available.
For all lines, must be correct to half square horizontal and vertical at the indicated vertices.*

$$x = 100, y = 200$$

B1

$y = x$ line
through (100, 100) and (200, 200)

B1

$2x + y = 600$ line
through (100, 400) and (200, 200)

B1

$3x + 2y = 800$ line
through (100, 250) and (200, 100)

B1

Feasible Region, all lines correct and region labelled
(condone no shading, ignore 'poor' shading)

B1

5

(iii) (Max) $y + \frac{3}{2}x$
PI by objective line with gradient -1.5

M1

$$(\quad = 400 + 150) = 550$$

A1

2

(iv) Buys 100 soft
400 medium
50 firm

B1

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1

[14]

Q7.

(a) $6x + 9y + 9z \leq 600$

Any of the three inequalities correct
(un)simplified, condone strict inequalities

M1

$$2x + 3y + 3z \leq 200$$

CAO

A1

$$9x + 6y + 9z \leq 600$$

$$3x + 2y + 3z \leq 200$$

CAO

A1

$$6x + 12y + 18z \geq 480$$

$$x + 2y + 3z \geq 80$$

CAO

A1

4

- (b) (i) $(z = y)$
 $2x + 3y + 3y \leq 200$ or $2x + 6y \leq 200$
*Correctly substitute into **this** inequality-
either simplified or unsimplified form*

M1

$$x + 3y \leq 100 \quad \text{AG}$$

$$3x + 2y + 3z \leq 200$$

*Correctly substitute into **this** inequality-
either simplified or unsimplified form*

$$(\Rightarrow) 3x + 5y \leq 200 \quad \text{AG}$$

$$x + 2y + 3z \geq 80$$

*Correctly substitute into **this** inequality-
either simplified or unsimplified form*

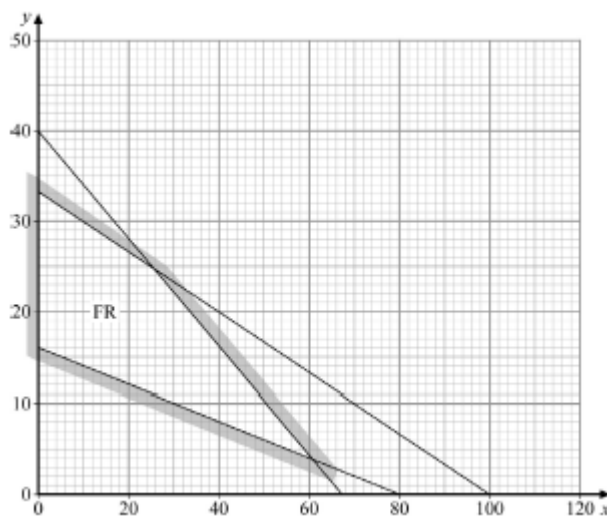
$$(\Rightarrow) x + 5y \geq 80 \quad \text{AG}$$

*All correct – must link their original inequality
to the stated answers*

A1

2

- (ii) Each line must be straight to have the B mark available.
For all lines, must be correct to $\frac{1}{2}$ square horizontal and
vertical at the indicated vertices.





Line through (10, 30) and (40, 20)

B1

Line through (50, 10) and (0, 40)

B1

Line through (80, 0) and (0, 16)

B1

FR, must have all lines correct and labelled
region (condone no shading)

B1

4

(iii) Max $x + 2y$

PI

If no statement (PI), then check OL on diagram,
which must be correct for M1

M1

Max (= 25 + 50) = 75

Note: 75 with no working 2/2

A1

2

(iv) 25 basic, 25 standard, 25 luxury

Condone "25 of each type" ONLY if (b)(iii)
fully correct

Note $x = 25 = y = z$ B0

B1F

1

EXAM PAPERS PRACTICE [13]

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Q8.

(a) $x + 5y \geq 25$ OE

ISW

B1

$2x + 15y \geq 60$ OE

ISW

B1

$x + 25y \geq 40$ OE

ISW

B1

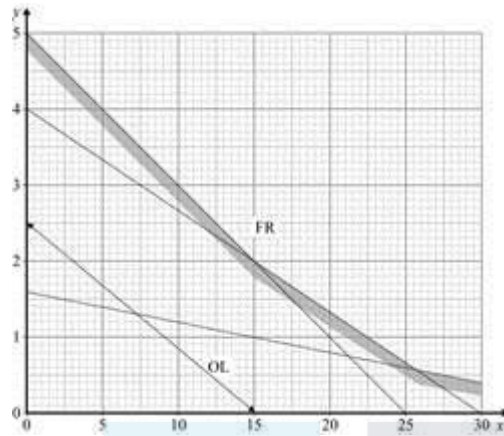
(C =) $2.5x + 15y$

ISW; condone $250x + 1500y$, but not any other multiples

B1

4

(b) (i)



Note: all points need to be correct to within half a square horizontally and vertically

Line through (0, 5) and (25, 0)

B1

Line through (0, 4) and (30, 0)

B1

Line through (15, 1) and (30, 0.4)

B1

FR, must have all lines correct and labelled region (condone no shading)

B1

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Objective line drawn, gradient of $-\frac{1}{6}$ or -6

M1

$$\text{Gradient} = -\frac{1}{6}$$

A1

6

(ii) 15 DIY, 2 trade

B1

1

(iii) (Cost) £67.50

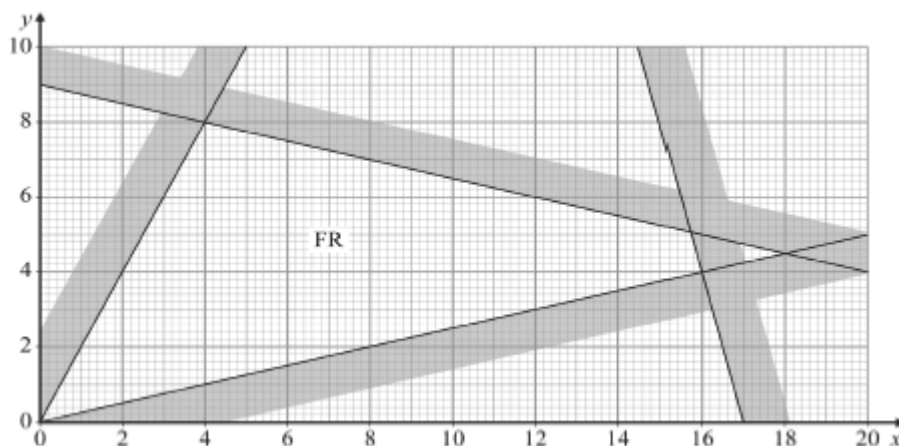
Condone 6750p, £67.5

B1

1

Q9.

(a)



line $y = mx$, must be correct to 1 square horizontally or vertically at origin

M1

through $(0, 0)$ and $(4, 8)$

A1

through $(0, 0)$ and $(16, 4)$

A1

line through $(15, 8)$ and $(17, 0)$

B1

line through $(4, 8)$ and $(12, 6)$

B1

FR must have scored previous 5 marks and labelled region (condone no shading)

B1

6

(b) (i) Max $(4, 8)$

Coordinates must be stated explicitly

B1

= 44

B1

2

(ii) Max $(16, 4)$

Coordinates must be stated explicitly

B1

= 84

B1

2

[10]

Q10.

(a) $x \geq 190, \quad y \geq 50, \quad z \geq 50$ oe

B1

$x + y + z \geq 300$ oe

B1

$2.5x + 2y + 2z \leq 1000$ oe

B1

$(5x + 4y + 4z \leq 2000$

$x \geq \frac{60}{100}(x + y + z)$

oe

B1

Strict inequalities: penalise first two instances only

$(2x \geq 3y + 3z)$

4

(b) (i) $y = z$
 $x \geq 190, \quad y \geq 50$
 $x + 2y \geq 300$

oe

$x + y = y \geq 300$ or $5x + 4y + 4y \leq 2000$

or $2x \geq 3y + 3y$

ie at least one clear line of working

showing substitution of $y = z$

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M1

$5x + 8y \leq 2000$

$2x \geq 6y$

$\left(y \leq \frac{1}{3}x \right)$

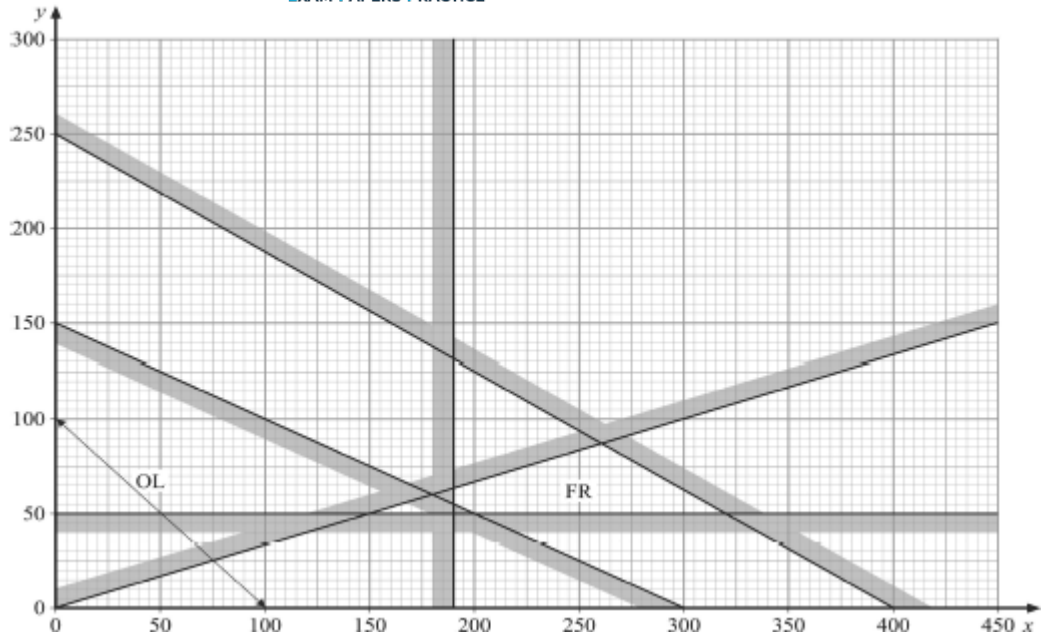
oe

AG All correct (3 'or' become 'and')

A1

2

(ii)



For all lines must be correct to $\frac{1}{2}$ square
horizontal or vertical

$$x = 190, y = 50$$

B1

through (0,150) and (300,0)

B1

through (0,250) and (400,0)

B1

$$y = mx \text{ through } (0,0)$$

M1

through (300,100)

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A1

Region must have all lines correct and
labelled region (condone lack of shading)

B1

A correct objective line

B1

7

(iii)
$$P = \frac{1}{2}x + \frac{1}{4}y + \frac{1}{4}z \quad \text{or} \quad \frac{1}{2}x + \frac{1}{2}y$$

PI

M1

Max at (320, 50)

B1

Profit $(160 + 25) = £185$

A1

Buys 320 slow, 50 medium, 50 fast

B1

*Note: (with no working) £185 3/4
320 slow, 50 medium, 50 fast 2/4
320 slow, 50 medium, 50 fast and £185 4/4*

4

[17]

Q11.

(a) $x + y + z \geq 110$

–1 for strict inequalities (max)

–1 for using g, p, s instead of x, y, z

(max)

$y \geq x$

B1

$y + z \leq 150$

B1

$16x + 8y + 24z \leq 3120$ ISW

$(2x + y + 3z \leq 390)$

B1

B1

$(P \Rightarrow) 70x + 30y + 50z$

B1

5

(b) (i) $z = 30$

Justify by correctly substituting into at least one of their inequalities

M1

$x + y \geq 80$ (or $x + y + 30 \geq 110$)

$(y \geq x)$

$y \leq 120$ (or $y + 30 \leq 150$)

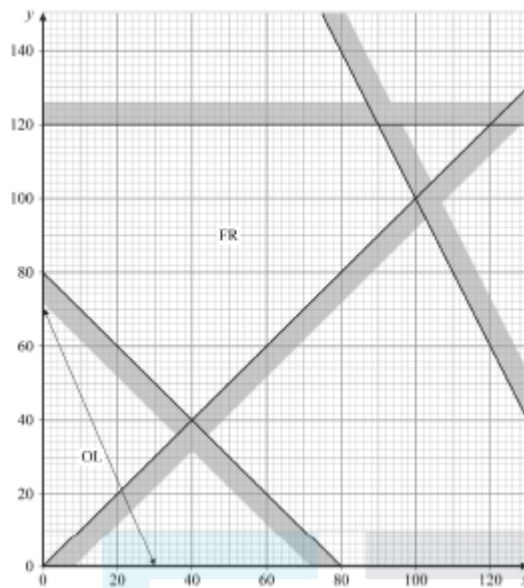
$2x + y \leq 300$ (or $2x + y + 90 \leq 390$ OE)

Correctly substituting into all 3 inequalities

A1

$(P = 70x + 30y + 1500)$

(ii)



$$y = 120$$

B1

$$x + y = 80$$

B1

$$y = x \text{ correct at } (40, 40) \text{ and } (100, 100)$$

B1

$$2x + y = 300, -ve \text{ gradient with one correct point in the interval } 80 \leq x \leq 120$$

M1

$$\text{Correct at } (100, 100) \text{ and } (90, 120)$$

A1

Correct region labelled

B1

$$OL: \text{ gradient of } -\frac{7}{3} \text{ or } -\frac{3}{7}$$

M1

$$\text{Gradient} = -\frac{7}{3}$$

A1

(iii) Considering (90, 120) and/or (100, 100)
Ignore other points being considered

M1

(£) 11500

A1

100 goats, 100 pigs, 30 sheep
CAO

A1

3

[18]

Q12.

Working must be in x , y and z
Equalities can only score M marks
Strict inequalities: -1 first error only

(a) $6x + 4y + 2z \leq 240$

M1

$3x + 2y + z \leq 120$
CAO

A1

$6x + 3y + 9z \leq 300$

M1

$2x + y + 3z \leq 100$
CAO

A1

$12x + 18y + 6z \leq 900$

M1

$2x + 3y + z \leq 150$
CAO

A1

$12x + 18y + 6z \geq 2(6x + 3y + 9z)$
OE

M1

$y \geq z$

CSO; OE in simplified form eg $y - z \geq 0$

A1

8

- (b) (i) $(z = x)$
 $4x + 2y \leq 120$ OE or $3x + 3y \leq 150$ OE
 $4x + 2y \leq 120$ OE $\Rightarrow 2x + y \leq 60$ **AG**
 $3x + 3y \leq 150$ OE $\Rightarrow x + y \leq 50$ **AG**
 $5x + y \leq 100, y \geq x$ **AG**

*Correct unsimplified subst $x = z$ into
either of these 2 correct inequs. (seen)*

M1

Both correct and simplified

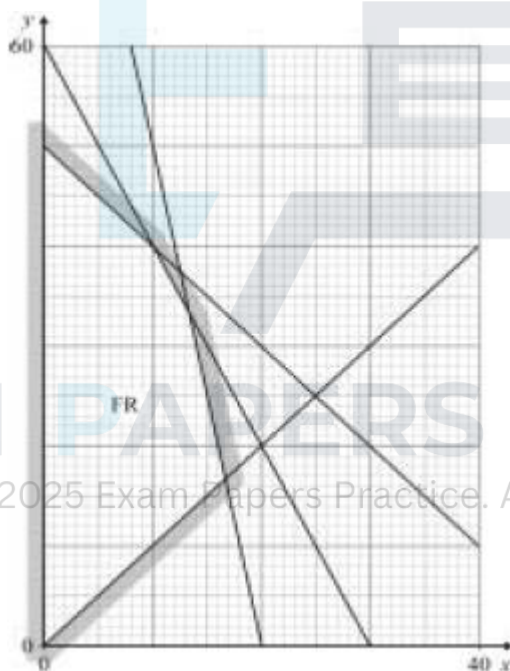
A1

Correct subst $x = z$ into 4 correct inequs.

A1

3

(ii)



Line 1 correct at (0, 50) (25, 25)

B1

Line 2 correct at (10, 50) (20, 0)

B1

Line 3 correct at (0, 60) (30, 0)

B1

Line 4 correct at (0, 0) (25, 25)

B1

Each line correct to $\frac{1}{2}$ square,

horizontally or vertically

FR, must have all lines correct **and**
labelled region (condone no shading)

B1

5

(iii) $N = x + y + z = 2x + y$

Stated or PI

M1

Max = 60

CSO; **SC** unsupported 60 scores 2/2

A1

2

- (iv) 10, 40, 10
11, 38, 11
12, 36, 12
13, 34, 13

Any correct; may be earned in part (iii)

3 correct

B1

B1

4 correct **and** no extras

B1

3

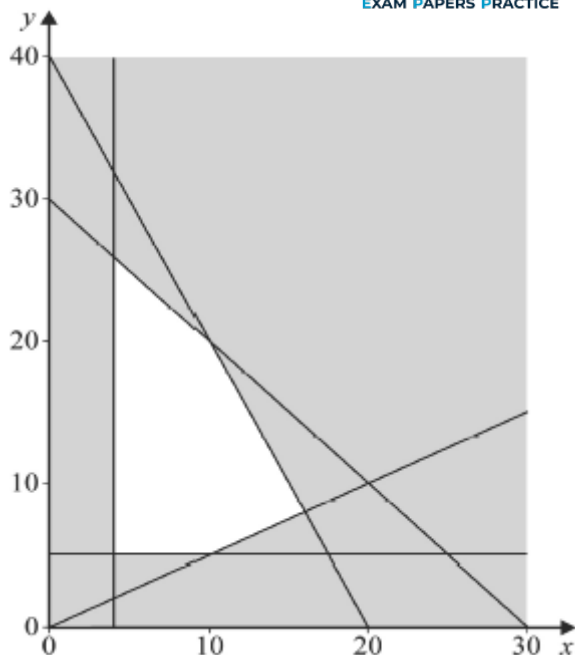
[21]

EXAM PAPERS PRACTICE

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Q13.

(a)



$$y = 5, x = 4$$

B1

$$x + y = 30$$

B1

$$2x + y = 40$$

B1

$$y = \frac{1}{2}x$$

B1

feasible region CAO

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B1

5

(b) (i) Max at $(16, 8) = 56$

Extreme point within $\frac{1}{2}$ square of their region

M1A1

2

(ii) Max at $(4, 26) = 82$

Extreme point within $\frac{1}{2}$ square of their region

M1A1

2

[9]

Q14.

$$\left. \begin{array}{l} 2x + 4y + 3z \leq 360 \\ 3x + 2y + 4z \leq 270 \\ x + 3y + 5z \leq 450 \end{array} \right\}$$

Any correct LHS in inequality

M1

OE

A2,1,0

$$6x + 9y + 12z \geq 720$$

M1

$$\Rightarrow x + 3y + 4z \geq 240$$

Allow further correct simplification

A1

$$2x + 4y + 3z \geq \frac{2}{5}(6x + 9y + 12z)$$

Must have 3 parts correct

M1A1

$$2y \geq 2x + 9z \quad \text{OE}$$

Allow further correct simplification

A1

8 [8]

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Q15.

(a) **All inequalities must be as below**

$$x \leq 100, y \leq 80$$

Both

B1

$$x + y \geq 60$$

B1

$$x < y$$

B1

$$2x + 8y \geq 320$$

OE

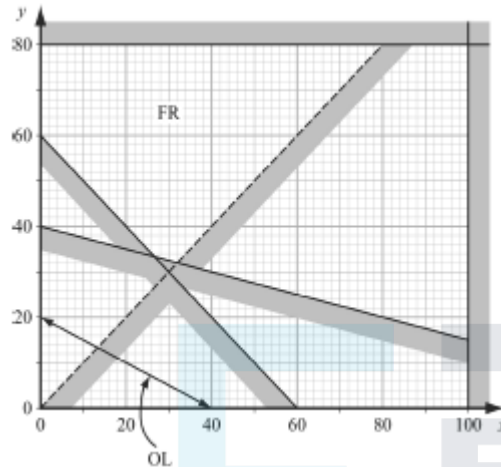
B1

(minimise $C \Rightarrow 1.5x + 3y$)

B1

5

(b)



$x = 100, y = 80$
within $\frac{1}{2}$ square
from $(0, 0)$ to $(80, 80)$

Other lines
within $\frac{1}{2}$ square
from $(0, 0)$ to $(80, 80)$

B1

B1 $\times$ 3

Feasible Region CAO (must have scored
B4 for drawing lines)
(condone $x = y$ as solid line)

B1

An Objective Line with gradient -0.5

B1

6

(c) Considering an extreme point in their region

M1

Min at intersect of $x + y = 60$

$$x + 4y = 160$$

PI by indication on diagram or

$$x = 26\frac{2}{3} \quad y = 33\frac{1}{3}$$

A1

Considering a pair of integer values where
 $26 \leq x \leq 28$, $32 \leq y \leq 34$

M1

(C =) £141 at (26, 34)
 or £141 at (28, 33)

A1

4

[15]

Q16.

- (a) $1000x + 500y \leq 9000$
 $(2x + y \leq 18)$

B1

1

- (b) $x \geq 2$, $y \geq 5$

B1

$y \geq 2x$

B1

$y \leq 3x$

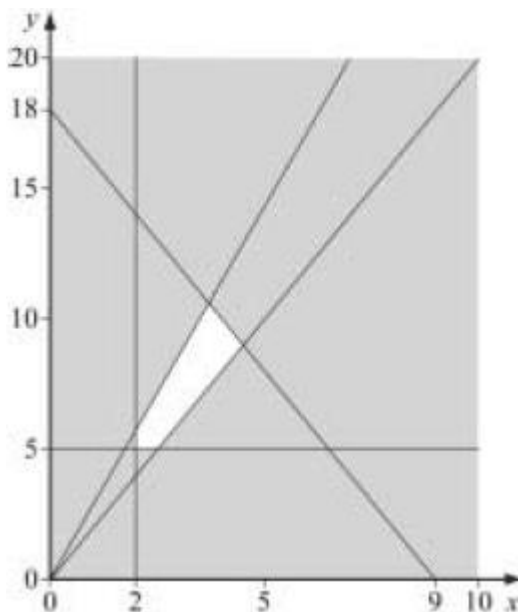
B1

-1 for strict inequalities
 -1 for 'w's and 'l's

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3

- (c)



$$x = 2, y = 5$$

B1

$$2x + y = 18$$

B1

$$\text{Line } y = mx$$

M1

$$y = 2x$$

A1

$$y = 3x$$

A1

Feasible region

B1

6

- (d) Considering an extreme point on their f.r.

Extreme point - vertex

M1

$$x = 4.5$$

A1

$$y = 9$$

A1

3

[13]

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Q17.

- (a) $5x + 10y \leq 1500$ (balloons)
 $\Rightarrow x + 2y \leq 300$

E1

$$32x + 8y \leq 4000$$
 (sweets)

E1

$$\Rightarrow 4x + y \leq 500$$

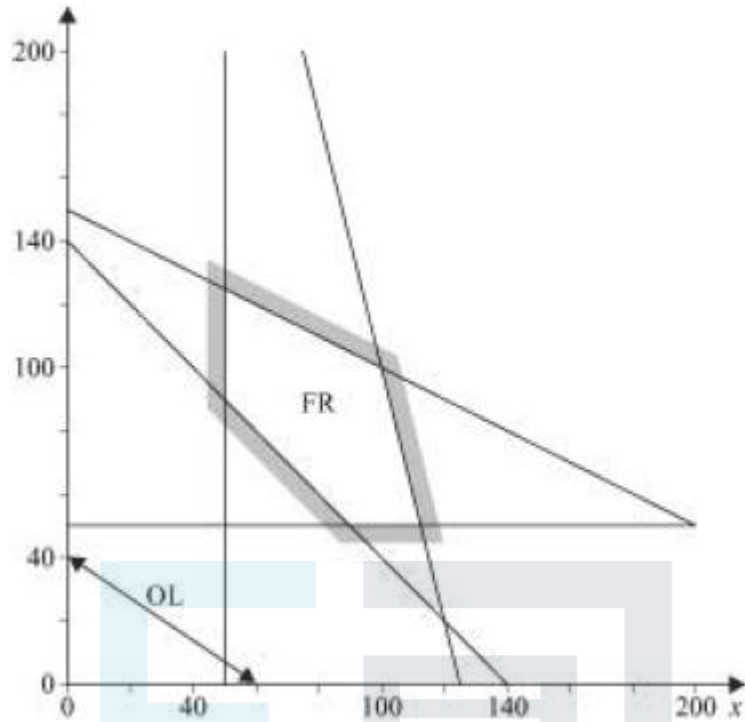
$x \geq 50, y \geq 50$, at least 50 of each

E1

$x + y \geq 140$, at least 140 in total

E1

(b) (i)



$x = 50, y = 50$

$x + y = 140$

B1

B1

Negative gradient (either)

M1

$4x + y = 500$

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A1

$x + 2y = 300$

A1

Feasible region

B1

Objective line drawn

M1A1

8

(ii) Maximum (100,100)

Considering extreme point on their region

M1

$= £200$

A1
2

- (iii) Minimum (90,50)
Considering extreme minimum point on their region

M1

$$= £132$$

A1
2

[16]

Q18.

- (a) (i) Max $2x + 3y$ at (30, 70)
Extreme

$$= 270$$

M1

A1
2

- (ii) Max $3x + 2y$ at (60, 40)
Extreme

$$= 260$$

M1

A1
2

- (iii) Min $-2x + y$ at (75, 10)
 $x = 75$

M1

$$= -140$$

A1
2

- (b) $x \geq 20, \quad y \geq 10$
OE

B1

$$x + y \leq 100$$

OE

B1

$2x + y \leq 160$ OE
 for gradient of -2

M1

A1

$y \leq x + 40$ OE
 for positive gradient

M1

A1

6

[12]

Q19.

(a) $10 \leq x \leq 80$

Strict inequalities -1 (or using p, c)

B1

$5 \leq y \leq 40$

B1

$x + y \leq 100$

$20x + 60y \leq 3000$ OE

B1

(maximise)(P =) $2x + y$

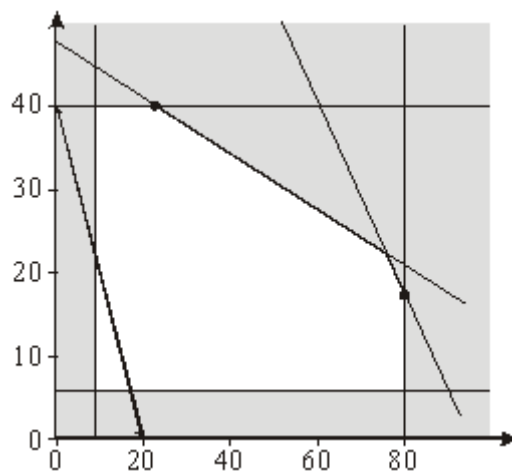
May be seen in (b) or (c)

B1

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5

(b)



For "x lines" and "y lines"

B1



$$\begin{cases} \text{For each other line M1 - ve gradient}(0, 50) \\ \text{M1 - ve gradient } (100, 0) \end{cases}$$

M1A1

M1A1

Feasible region correct to within 1 square

B1

Objective line

B1

7

(c) Max at (80,20)

Considering an extreme point in their region

M1

$$P = £180$$

region

A1

2

(d) $P = x + 4y$
Max at (30, 40)

M1

$$P = £190$$

Using (30,40) ($\pm$ square)

A1

2

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[16]

Q20.

(a) $4x + 2y \leq 5 \times 4 \times 60$

Condone =

B1

1

(b) $x \geq 40, y \geq 40$

Both

B1

$$x + y \geq 120$$

$$x + y \leq 400$$

Both

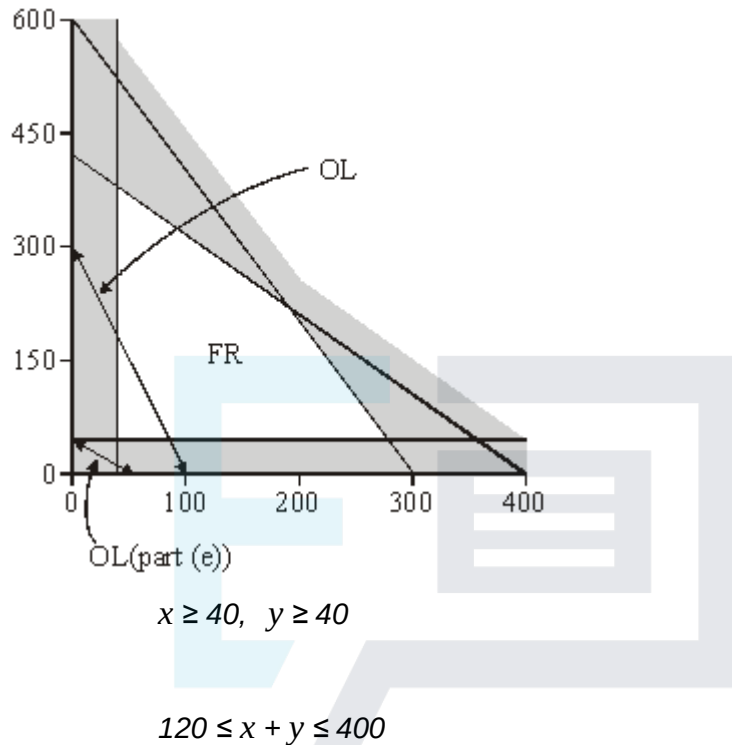
B1

$$(P =) 3x + y$$

B1

3

(c)



B1

B1

EXAM PAPERS PRACTICE

B1

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Correct FR

B1

Correct OL

B1

5

(d) Extreme points

M1

Max at $x = 280, y = 40$

A1

$$P = 840 + 40 = \text{£}880$$

SC: (280, 20) scores 1/3

B1

3

(e) (i) Max at (200, 200) → (40, 360)

M1

Profit £800

A1

2

(ii) No of combinations

B1

$$200 - 40 = 160$$

$$\begin{array}{r} + 1 \\ \hline 161 \end{array}$$

B1

2

[16]

Q21.

(a) Milky
 $12x + 18y \leq 600$
 $\Rightarrow 2x + 3y \leq 100$

B1

1

(b) $x \geq 15, y \geq 15$

B1

$$x + y \geq 35$$

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B1

$$20x + 10y \leq 600$$

OE

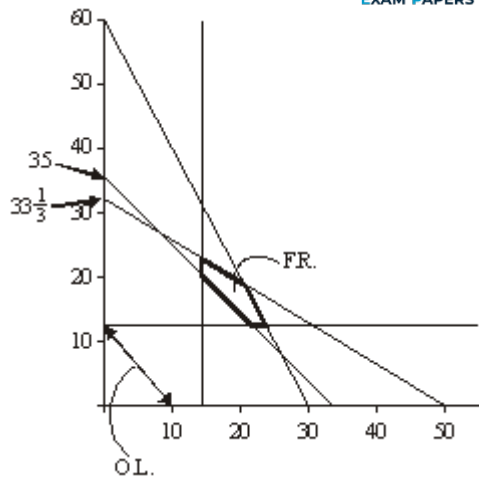
B1

$$(P =) 1.5x + y$$

B1

4

(c)



$$x \geq 15, y \geq 15$$

Other 3 lines

Feasible region

Objective Line

B1

B1 × 3

B1

B1

6

(d) Considering one of their extreme points

M1

$$P = 50$$

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A1

2

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[13]

Q22.

(a)

$$\left. \begin{array}{l} \text{At least 30 \& 60} \\ \text{In total } \leq 200 \end{array} \right\}$$

B1

$$\text{Area } 2x + 3y \geq 300$$

B1

$$\left. \begin{array}{l} \text{Cost } 40x + 12y \leq 3600 \\ 10x + 3y \leq 900 \end{array} \right\}$$

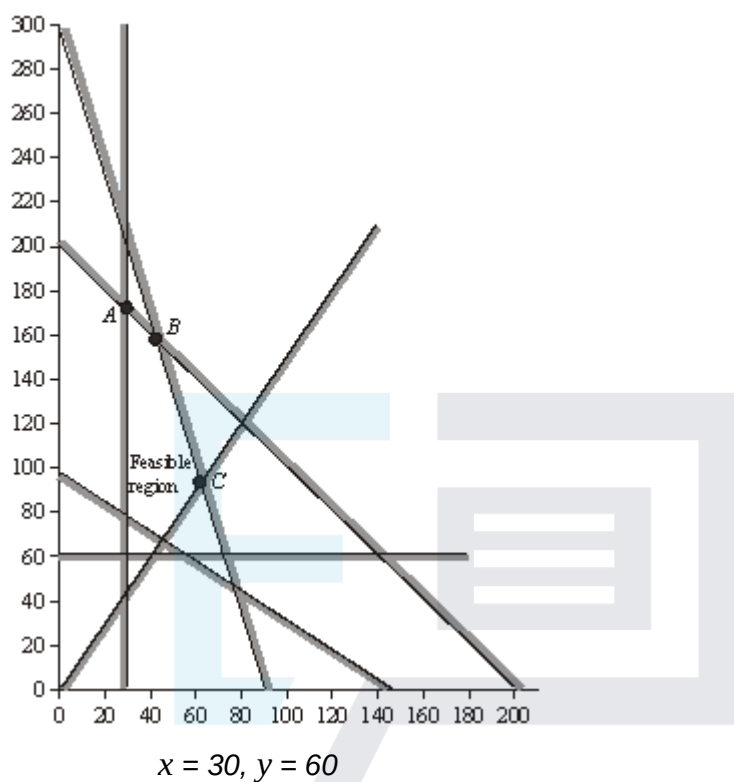
B1

y at least 150% x
(strict)

B1

4

(b)



B1

EXAM PAPERS PRACTICE

B1×4

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B1

6

(c) $y \geq 60$

B1

1

(d) $P = 4x + 5y$
Max at A

Considering extreme point(s)

M1

$$\left. \begin{array}{l} x = 30 \\ y = 170 \end{array} \right\}$$

A1

$P = 970$

B1

3

[14]

Q23.

(a) $y \geq \frac{1}{2}x \Rightarrow 2y \geq x$

B1

$20x + 10y \leq 600 \quad \text{so} \quad 2x + y \leq 60$

B1

2

(b) $x \geq 5, y \geq 5, y \leq 25$

all correct

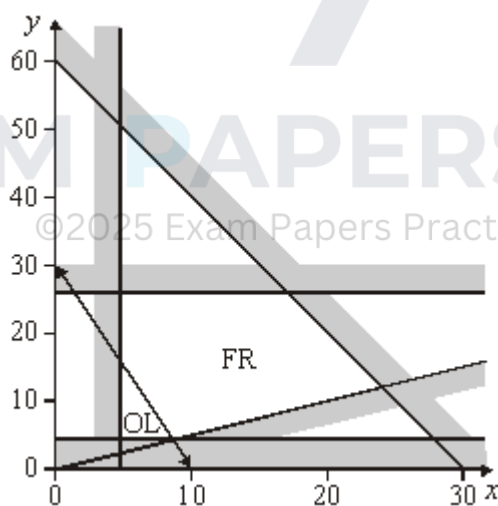
B1

$P = 3x + y$

B1

2

(c)



diagonal lines

B1 × 2

$x \geq 5, y \geq 5, y \leq 25$

B1

feasible region

B1

objective line

B1

5

(d) Max at $x = 24, y = 12$

M1

$$P = 84$$

A1

2

(e) $P = 2x + 2y$
Max at (17.5, 25)

M1

Not integers
Values (17, 25) and (18, 24)

B1,B1

give $P = 510$ (6 hrs) + 84 (1hr)
= £594

CAO

A1

4

[15]

Q24.

(a) $x \geq 2, y \geq 5, z \geq 2$

All

B1

$x + y + z \leq 28$

B1

$$40 \leq 3x + 2y + 2z \leq 60$$

B1+1

4

(b) (i) $x = z$

$$\therefore x \geq 2, y \geq 5$$

$$2x + y \leq 28$$

B1(*)

$$40 \leq 5x + 2y \leq 60$$

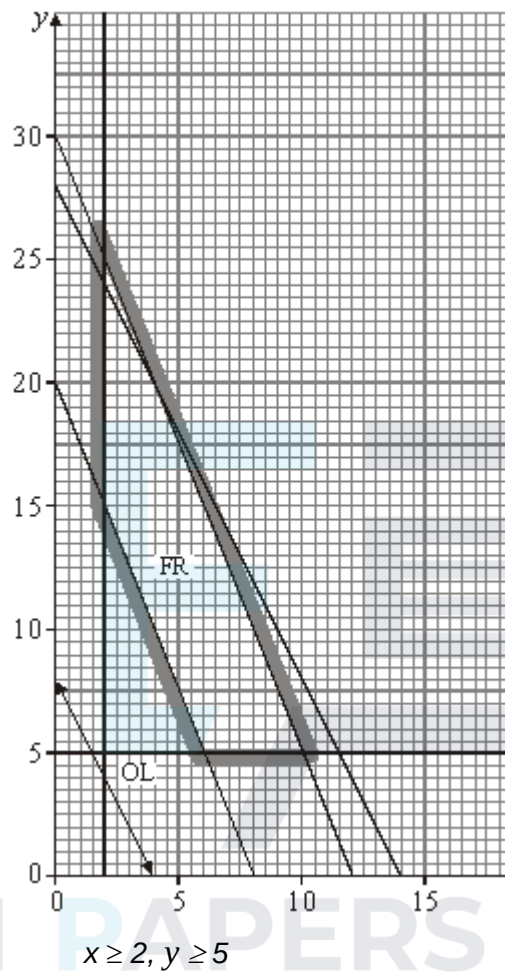
(*) Must have shown

$$x + z = 2x \text{ etc}$$

B1(*)

2

(ii)



$$x \geq 2, y \geq 5$$

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B1

other lines

B1 × 3

region (pentagon)

B1F

objective line

M1

Correct

A1

7

(iii) Iron 28

$$\therefore x = 2, y = 24$$

Attempt at a max extreme point on their graph

M1

or $x = 3, y = 22$
 or $x = 4, y = 20$
 any pair
 or 2,2,24
 3,3,22
 4,4,20

A1

2

[15]

Q25.

(a) $3x + 6y \leq 600$
 $\Rightarrow x + 2y \leq 200$

or roses

$6x + 3y \leq 600$
 $\Rightarrow 2x + y \leq 200$

or carnations

$4x + 4y \leq 480$

or lilies

$\Rightarrow x + y \leq 120$

OE

$x \geq 0, y \geq 0$

(condone absence)

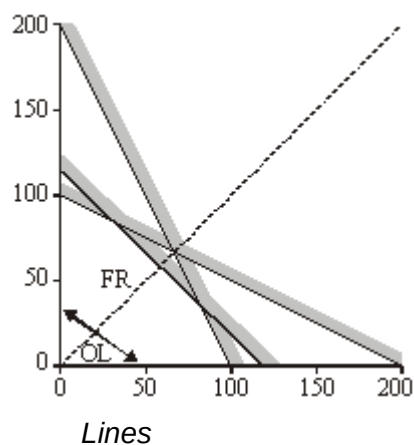
B1 $\times$ 3

3

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(b) $P = 1.5x + 2.5y$
 (maybe shown in part (c))

B1



B1 × 3

Region

B1

Objective line with -ve gradient

M1

A1

7

(c) Checking an extreme point

M1

$$P = 260$$

A1

2

(d) $x > y$

or $y = x$ on diagram

M1

Max at (60, 60)

Attempt at points around (60, 60)

M1

But $x > y$

$$\therefore x = 61, y = 59$$

A1

$$P = 239$$

A1

4

[16]

Q26.

(a) $x \geq 7, y \geq 7$

Both

B1

$$y \leq 2x$$

B1

$$y \geq \frac{1}{3}x$$

B1

$$2y + x \leq 40$$

OE; M1 for – ve gradient

M1A1

$$3x + 4y \geq 60$$

OE; M1 for – ve gradient

M1A1

7

(b) Max at (8, 16)

M1A1

$$= 96$$

M1 Considering extremes

A1

3

[10]

Q27.

(a) $x \geq 20, y \geq 20$

–1 for strict inequalities

B1

$$x + y \geq 70$$

B1

$$2x + y \geq 100$$

B1

$$3x + 2y \leq 300$$

B1

$$(N =) x + y$$

B1

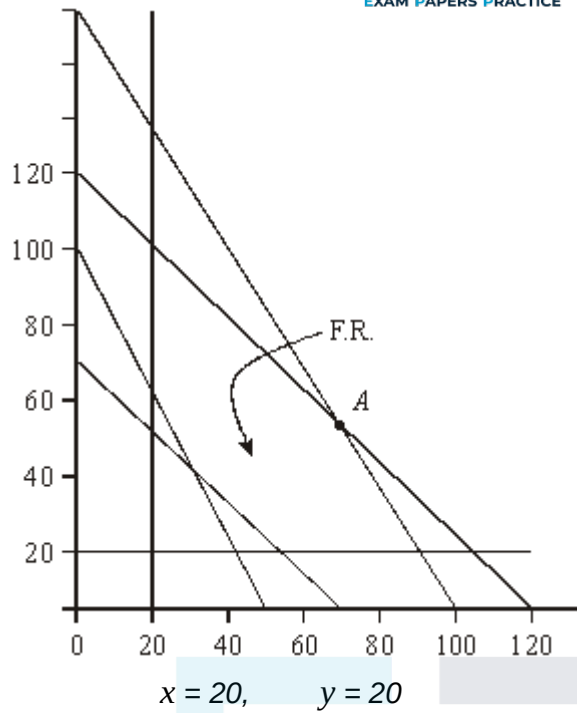
$$4x + 4y \leq 480$$

OE

B1

6

(b)



other lines

F.R.; Pl
ft only if FR is hexagonal

B1

B1 $\times$ 4

B1F

6

- (c) (i) Max no. is $x = 20, y = 100$
on their F.R.

M1

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Total = 120

A1

2

- (ii) Point A is (60, 60)
No. of sols is (20, 100) $\rightarrow$ (60, 60)
Considering a range of solutions in their F.R.

M1

= 41

SC: 40 scores 1

A1

2

[16]

Q28.

- (a) $x \geq 1, y \geq 1$
(-1 for > used)

B1

$6 \leq 3x + y \leq 12$
Condone using a and d

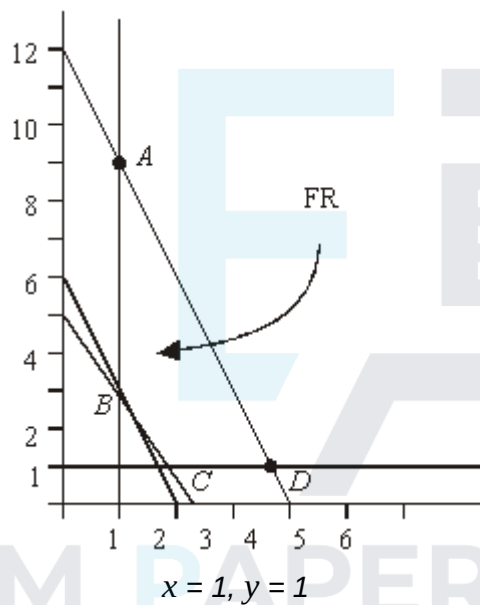
B1 + B1

$2x + y \geq 5$

B1

4

(b)



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 3 lines

B1 4 3

FR CAO

B1

5

- (c) (i) Max points at A
 $x = 1, y = 9$

M1

Points = $9 + 2 = 11$

A1

2

- (ii) Spending £11m

$$3x + y = 11$$

PI

M1

Either $(2, 5) \Rightarrow 9$ pts

$\left. \begin{array}{l} \text{Either } (2, 5) \Rightarrow 9 \text{ pts} \\ \text{or } (2, 3) \Rightarrow 8 \text{ pts} \end{array} \right\}$

Considering points in c's feasible region or c's line

M1

$$\therefore x = 1, y = 8$$

A1

3

[14]

Q29.

(a) $x \geq 2, y \geq 2$

Both()*

B1

$$x \leq 20, y \leq 20$$

Both()*

B1

$$x + y \geq 8$$

OE()*

B1

$$x + y \leq 24$$

OE()*

()Strict inequalities -1*

B1

4

(b) (i) Max at $(2, 20) (4, 20) (20, 4) (20, 2)$

M1

$$\therefore x + 3y = 64$$

A1

2

(ii) Min at (20, 2)

M1A1

$$\text{Min } \therefore -x + 2y = -16$$

A1

3

[9]

Q30.

- (a) (i) Extreme points
(0, 120), (50, 100), (100, 0)

M1

Values 360, 400, 200

*any of
or by drawing objective*

Maximum = 400
CAO

A1

A1

3

- (ii) Extreme points
(0, 50), (50, 0)

M1

2

Values 50, 200

or by drawing objective

Minimum = 50

A1

- (b) $x \geq 0, y \geq 0$

Allow strict inequalities

B1

6

$$x + y \geq 50$$

B1

$$2x + y \leq 200$$

for negative gradient

M1

OE

A1

$$2x + 5y \leq 600$$

for fractional gradient

OE

M1/A1

[11]

Q31.

(a) $2x + 4y \leq 50$

OE (Strict inequalities –1)

B1

$$3x + y \leq 24$$

(equalities – 1)

B1

$$x + y \leq 20$$

(A & B –1)

B1

$$x \geq 2, y \geq 2$$

Both

B1

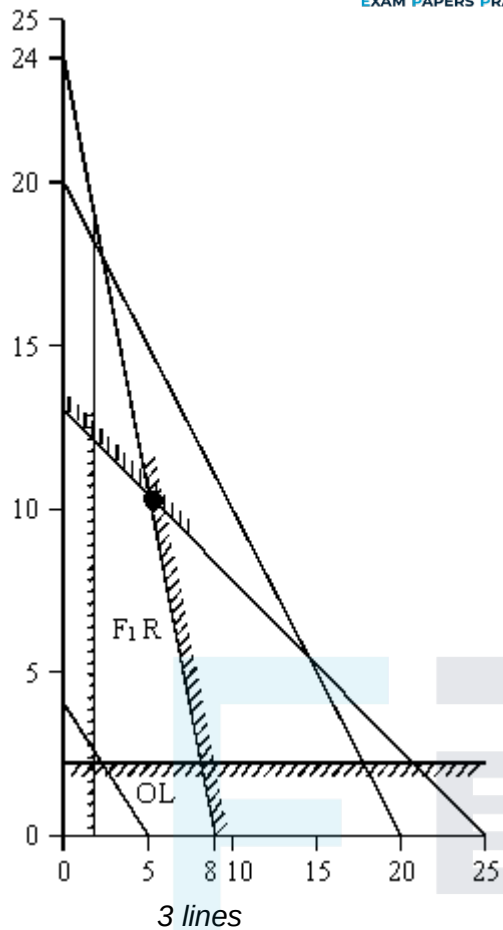
$$(T =)20x + 25y$$

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B1

5

(b)



B1 4 3

$x = 2, y = 2$ (Both)

B1

closed region marked

B1F

Objective line

B1

6

- (c) $T = 20x + 25y$
(Min at $(2, 2) =$)£90

M1

Considering extreme points on their region

A1

Max at $(4.6, 10.2)$

Considering $(3, 11)$ $(4, 10)$ $(5, 9)$

B2, 1

Impossible

Max = £335

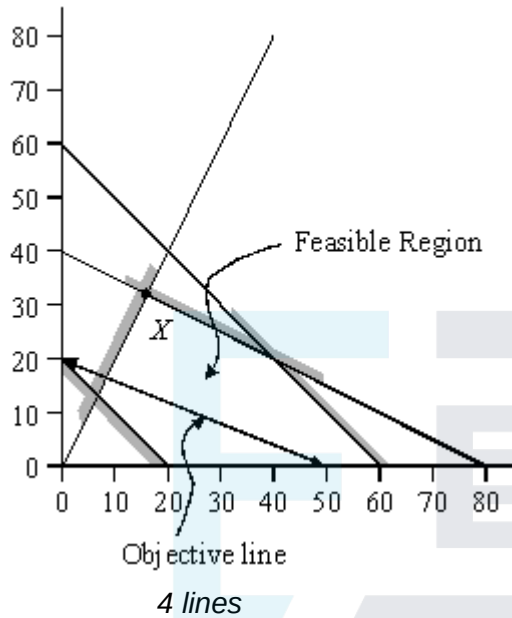
B2

6

[17]

Q32.

(a)



Region (must be pentagon)

B1 × 4

B1F

M1

Objective line

Correct

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A1

7

(b) Max value of F at X

$$x = 16, y = 32$$

Considering extreme points

M1

$$F = 32 + 160 = 192$$

A1

2

[9]

Q33.

(a) $8x + 8y \leq 7200$

$2x + 3y \leq 2200$

$x \geq 300, y \geq 300$

$x + y \geq 800$

$T = 20x + 25y$

–1 for any extra

–1 for strict inequalities

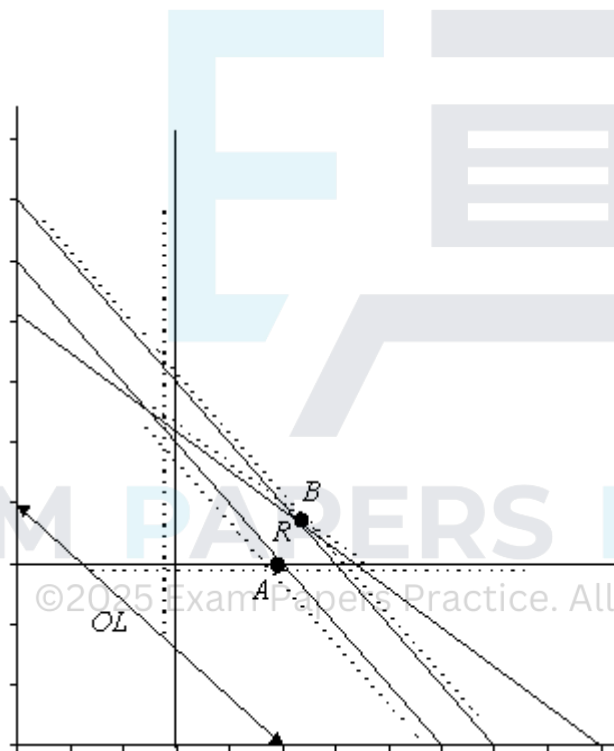
B1

B1

B1

B15

(b)



$x + y = 900$

B1

$2x + 3y = 2200$

B1

$x = 300 \text{ and } y = 300$

B1

$x + y = 800$

B1

region – pentagon

B1

Line

B1

6

- (c) Minimum T at A (500, 300)

M1

$$T = £17\,500$$

M1

Maximum I at B (500, 400)

$$I = £20\,000$$

c. a. o } method allowed
even if with
wrong graph
c. a. o }

M1

A1

4

[15]

Q34.

- (a) $x > y + z$

B1

$$y \leq \frac{15}{100}(x + y + z)$$

M1

$$\Rightarrow 17y \leq 3x + 3z$$

allow integer multiples

A1

$$y \geq z$$

B1

$$z < \frac{3}{5}x$$

M1

$$5z < 3x$$

as above

A1

$$(C \Rightarrow) 8x + 12y + 14z$$

B1

7

(b) $y = 6$
 $\Rightarrow x + z \geq 34$

attempt at substituting $y = 6$

M1

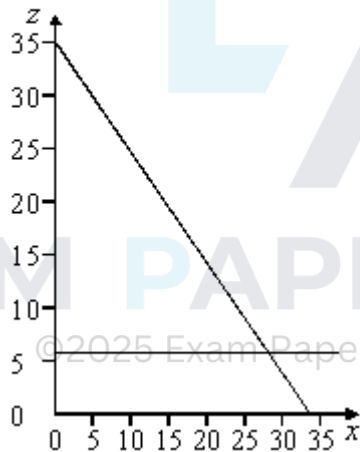
$$z \leq 6$$

$$3x > 5z$$

at least two correct

A1F

$$x > z + 6$$



one for each line

G2

$$x = 34, \quad z = 0$$

attempt at considering extreme points

M1

$$\text{Cost} = \text{£}344\,000$$

c.a.o

A1

6

Alternative:

$y = 6, z \leq 6$
 $\therefore x > 6 + z$
 $(17 \times 6) \leq 3x + 3z$
 $102 - 3z \leq 3x$
 $x > 10$
 $\Rightarrow x \geq 28$
 Minimum at $(38 \times 8) + (6 \times 12)$
attempt at points $(28, 6) \rightarrow (34, 0)$

B1

M1

A1

B1

M1

= £344 000

A1

[13]

Q35.

- (a) Permission $x + y \leq 16$
 $x + y \geq 12$

B1

Must build $x \geq 4, y \geq 4$

B1

and $22500x + 30000y \leq 450000$

Method for land or build

M1

$3x + 4y \leq 60$

Build $20000x + 15000y \leq 300000$

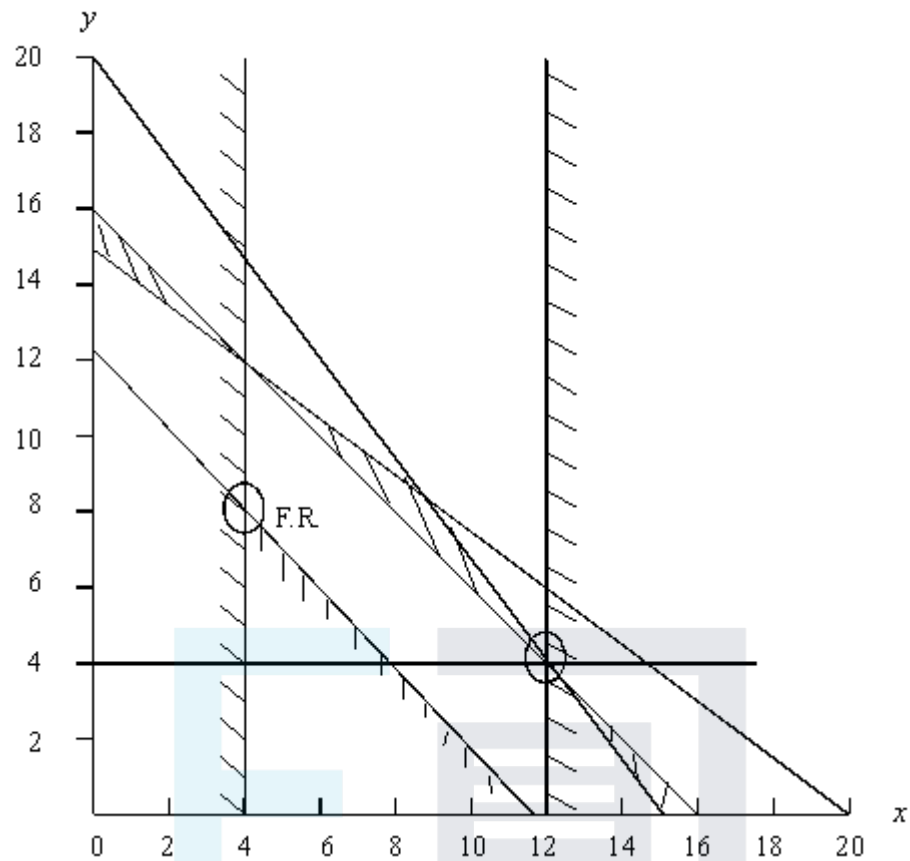
A1

$4x + 3y \leq 60$

A1

5

(b)



$$12 = x + y = 16$$

B1

$$x = 4, y = 4$$

B1

$$3x + 4y = 60$$

B1

$$4x + 3y = 60$$

B1

Region

B1

5

- (c) Profit line $x + y = P$
Points (4, 8) (5, 7) (6, 6) (7, 5) (8, 4)

B1

Profit £120, 000

B1

2

(d) $190000 = x(s - 42500)$

M1

$+ y(s - 45000)$

A1

Using (4, 8) (12, 4)

B1

$(4, 8) =) S = £60,000$

B1

$(12, 4) =) S = £55,000$

A1

A1

6

[18]

Q36.

(a) $x \geq 5, y \geq 20$

Both

B1

$3x + 3.5y \leq 210$

B1

$6x + 2y \leq 180$

Or equivalent

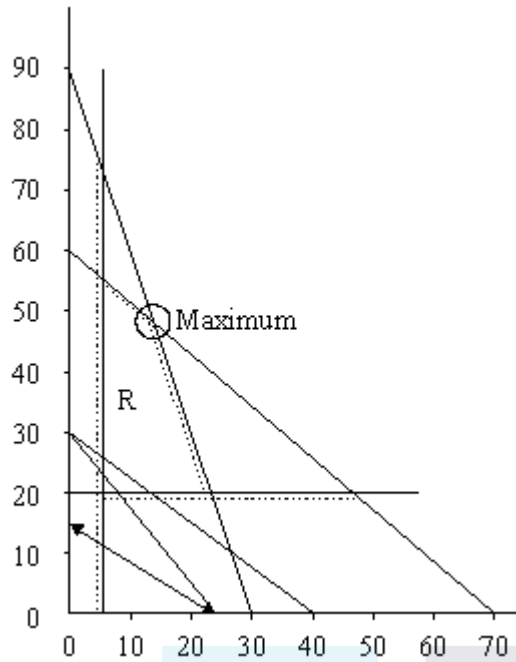
B1

$30x + 40y \geq 1200$

B1

4

(b)



Scale

$x \geq 5, y \geq 20$: both:

Other 3 lines (-1 EE)

Region correct and labelled

B1F

B1F

B2F

B1F

5

- (c) (i) Line parallel to $6x + 5y = 150$

An objective line

M1

Correct objective line

A1

2

- (ii) Maximum at (14, 48)

M1

Pay = $42 + 120 = £162$

A1

2

- (d) Line parallel to $5x + 8y = 120$

M1

Maximum at (5, 55.7) – IMP

M1

$(5, 55) \Rightarrow \text{£}232.50$

Must include impossibility to score full marks

A1

$(6, 54) \Rightarrow \text{£}231$

A1

4

[17]

Q37.

(a) $x \geq 0, y \geq 0$

Must be at least 0, real life

B1

A : $90x + 90y \leq 1200$

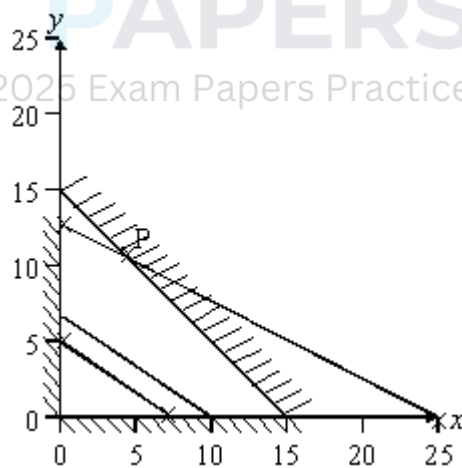
B1

B : $60x + 120y \leq 1500$

B1

3

(b) (i)



suitable scale

B1

2 lines

B2

Region

	Excluding negatives	B1	
		B1	5
(ii)	Objective line	M1A1	
	Vertex P	B1	3
(iii)	Coords of P $(1\frac{2}{3}, 11\frac{2}{3})$ must be integers	E1	1
(iv)	$P = 10x + 12y$ $(1, 11) \quad P = 142$ $(1, 12) \quad P = 154$ <i>considering any part</i> $(2, 11) \quad P = 152$ $(2, 12) \quad \text{Not in region}$ Hence $x = 1, y = 12$ $P = 154$	A1	

A1cao

3

[15]

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Q38.

(a) $6x + 4y + 2z \leq 240$

M1

$3x + 2y + z \leq 120$

A1

$30x + 40y + 60z \leq 3300$

$3x + 4y + 6z \leq 330$

A1

3

(b) (i) $y = z$
 $3x + 3y \leq 120$

M1

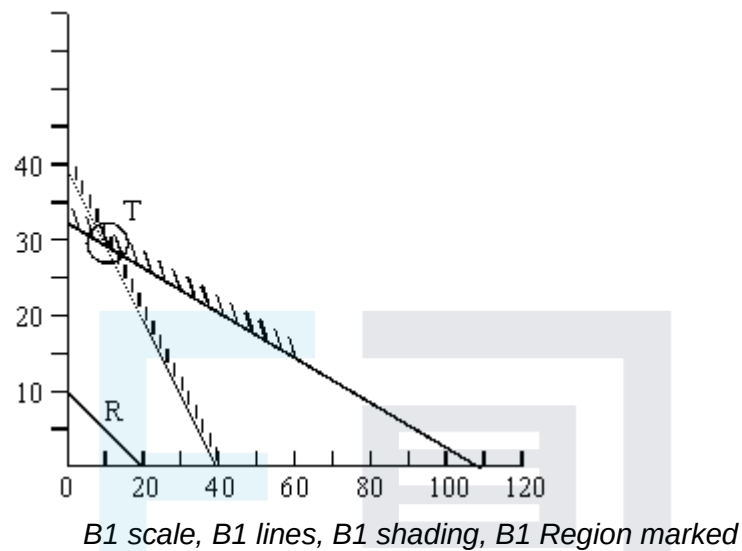
$$\Rightarrow x + y \leq 40$$

$$3x + 10y \leq 330$$

Both

A1

(ii)



B1B1B1B1

4

(iii) Object line shown
T shown

3

(iv) $x = 10, y = 30$

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1

$$T = 70$$

(c) $3x + 10y \leq 288$

B1

4

Intersect at (16, 24)

M1A1

Max is 64

B1

[17]

Q39.

- (a) (i) Time : $60x + 20y \leq 900$
 $\Rightarrow 3x + y \leq 45$
using time

B1

- (ii) $x \geq 3, y \geq 3$
 $600x + 120y \geq 4800$

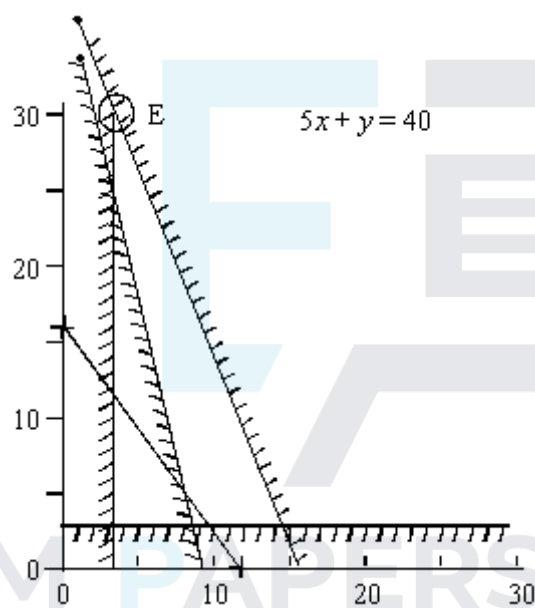
B1+B1

or in simplified form

B1

4

- (b) (i)



suitable scales

B1

$x = 3, y = 3$

B1

$3x + y = 45$

B1F

$5x + y = 40$

B1F

Correct region (quadrilateral)

B1F

5

- (ii) Line $4x + 3y = 48$

B1

E marked

		B1	2
(c)	(i) $x = 3, y = 36$ <i>from quad</i>		
		B1F+B1F	2
	(ii) $(48x + 36y) \times 45$ $= £64800$		
		B1F	1
(d)	(i) $90x + 18y = 1170$ $\Rightarrow x = 10 \quad y = 15$ or $x = 11 \quad y = 10$ $x = 12 \quad y = 5$ <i>Any correct pair</i>	M1 A1cao	2
	(ii) MAX $x = 14 \quad y = 3$ Earnings = 1314	B1F+B1F	
		B1cao	3

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[19]

Q40.

- (a) cost
 $12500x + 7500y \leq 187500$
 $\Rightarrow 5x + y \leq 75$
Equalities missing
- Insurance
 $\Rightarrow x + 2y \leq 29$
penalise by – 1 once
- Demand
 $\Rightarrow x + y \geq 10$

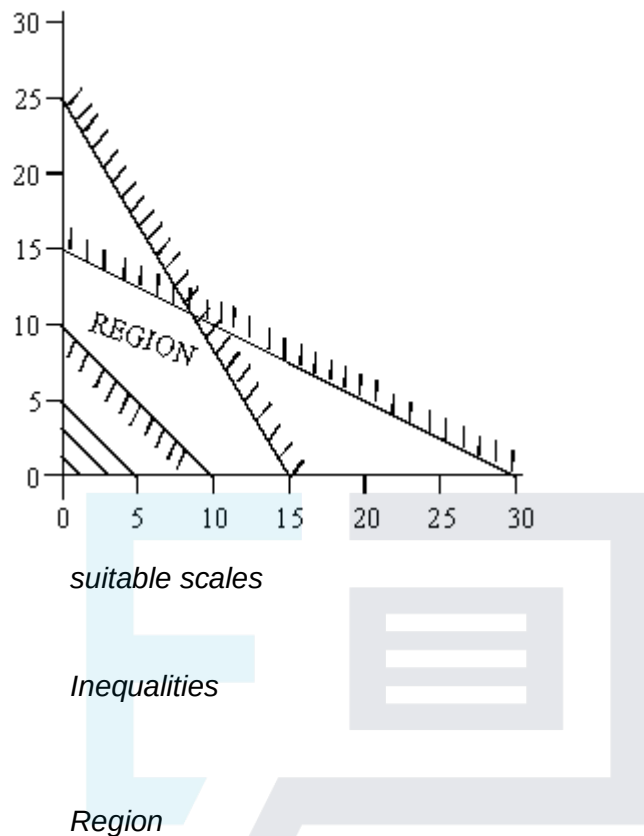
e1

E1

E1

3

(b) (i)



B1

B143

5

B1

(ii) Profit line

M1A1

Vertex

B1

3

(iii) $x = 9, y = 10$

B1

$$P = (9 \times 1200) + (10 \times 1200)$$

M1

$$= £22800$$

A1F

3

[14]

Q41.

- (a) x necklaces a week at one per working day $\Rightarrow 0 \leq x \leq 7$ (similarly for y)

B1

y bracelets only follow some of the necklaces, so $y \leq x$

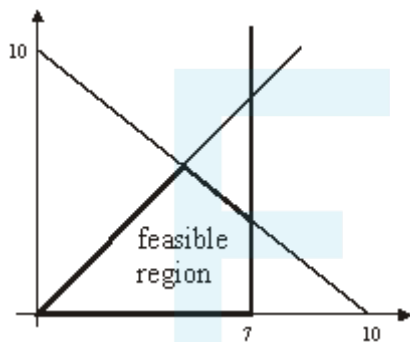
B1

We also need $x + y \leq 10$.

B1

3

(b)



One per boundary line (inc ft on third)

Region

B1 B1 B1ft

B1

4

- (c) (i) Vertices of feasible region
(or use 'profit line')

M1

are (0, 0), (5, 5), (7, 3) and (7, 0)

A1

Income of $20x + 10y$ is maximised at (7, 3) so he should make 7 necklaces and

A1

3 bracelets each week.

B1

4

- (ii) With the bracelets costing £B we want the maximum of $20x + By$ to be attained at

M1

(5, 5).

If $B < 20$ then the maximum is attained at

A1

(7, 3), but if $B \geq 20$ then the maximum is

M1

attained at (5, 5) as required. So must charge at least £20.

A1

4

[15]

Q42.

(a) Flour $\Rightarrow 150x + 300y \leq 7500$
 $\Rightarrow x + 2y \leq 50$

B1

Butter $\Rightarrow 5(0)x + 2(0)y \leq 100(0)$

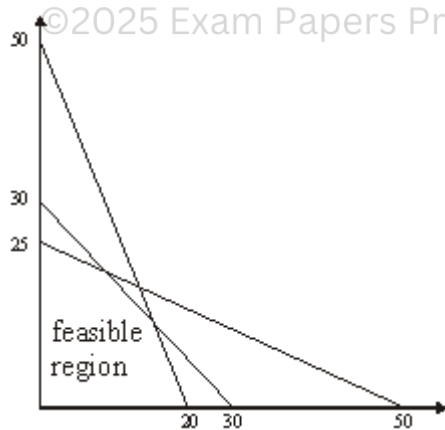
B1

Eggs $\Rightarrow x + y \leq 30$

B1

3

(b)



B1ft

One per line (ft)

B1ft

Region (ft)

B1ft

4

The feasible region has vertices
(or by profit lines)

M1

(0,0) (0,25) (10,20) ($13\frac{1}{3}$, $16\frac{2}{3}$) (20,0)

A1

The profit of $x + 1\frac{1}{2}y$ is maximised at
(10,20) so they should make 10 Romanos

A1

and 20 Sardinios.

B1

4

(c) With profit $2x + 1\frac{1}{2}y$ the maximum is

M1

reached at ($13\frac{1}{3}$, $16\frac{2}{3}$) but it is
impractical to make a fraction of a pizza.

A1

(d) Searching integer points nearby gives

M1

(13,17) as the best in the region, so make
13 Romanos and 17 Sardinios in this case.

A1

4

[15]

Q43.

(a) enough nails $\Leftrightarrow 1000x + 100y \geq 5000$

B1

enough screws $\Leftrightarrow 1500x + 100y \geq 60$

A1

$$\Leftrightarrow 15x + y \geq 60$$

A1

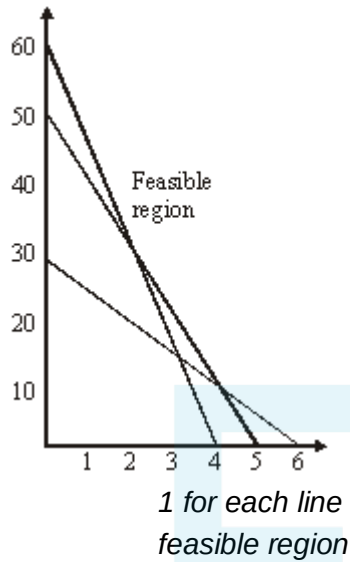
enough nails $\Leftrightarrow 500x + 100y \geq 3000$

$$\Leftrightarrow 5x + y \geq 30$$

A1

4

(b)



B1 B1 B1 M1 A1

5

(c) $C = 6x + y$ is minimised at (4,10)

Vertex check or lines/gradients method

M1

(with a value of 34)

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A1 A1

Should buy 4X-packs & 10 Y-packs

sc B2 for cao

B1

4

(d) With '3 for the price of 2' again a vertex

(strictly "2/3 price" for linearity: but this does yield multiples of 3)

M1

check gives minimum at (6,0) (of £24) so buy 6 X-packs only.

A1 B1

3

[16]

Q44.

- (a) Line 1: $3x + 2y = 42$
 Line 2: $x + 2y = 30$
 Line 3: $x + y = 16$
For one

B1

For other two

B1

2

- (b) $3x + 2y = 42$, $x + y = 16 \Rightarrow$

B1

$x = 10, y = 6$

For either co-ordinate

A1

Hence C is $(10,6)$

(or by lines/gradients)

A1

3

- (c) Trying all vertices leads to

M1

$P = 2x + 3y$ maximised at $(2,14)$

A1A1

So maximum of P is 46 by making
 2 Xtremes and 14 Yltras

A1

4

- (d) (i) New constraint is $y \leq 0.2(x + y)$ and so

M1

$4y \leq x.$

A1

2

- (ii) This crosses the boundary of the feasible

M1

region at $(12,3)$.

A1

In new region maximum of P is at $(12,3)$

M1

so they should make 12 Xtremes and 3 Yltras

A1

4

[15]

Q45.

- (a) x Xtravim give $3x$ mg of vitamin A
 y Yeasty give y mg of vitamin A
 Therefore we need $3x + y \geq 15$.

B1

Similarly B $\Rightarrow x + 4y \geq 20$.

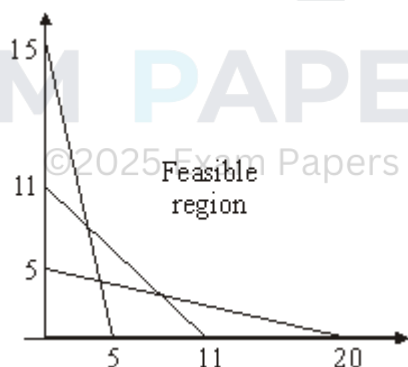
B1

and C $\Rightarrow x + y \geq 11$

B1

3

(b)



*B1f.t B1f.t B1f.t} One per line (ft)
 Correct region marked*

B1

4

- (c) (i) Vertices of feasible region are
 $(0, 15)$, $(2, 9)$, $(8, 3)$ and $(20, 0)$

M1

(or use a cost line)

A1

Cost of $2x + 5y$ is minimised at $(8, 3)$ so he should take 8 Xtravim and 3 Yeasty each day.

A1

B1

4

- (ii) The minimum cost is attained at two vertices, one with y-coordinate 0.

M1

A1

So the cost line must be parallel to the line from $(8, 3)$ to $(20, 0)$ (i.e. $x + 4y$).

M1

So the Xtravim tablets cost one quarter of the Yeasty; i.e. 1.25 p each.

A1

4

[15]

Q46.

- (a) Number by sea = $1000 - x - y$

B1

$$100 \leq 1000 - x - y \leq 500$$

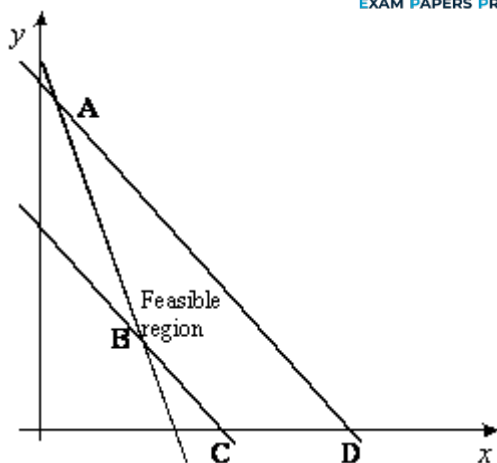
B1

$$1000 - y \leq 2x$$

B1

3

- (b)



Axes

M1

A1



A1

A1

A1

5

(c) (i) Total cost = $50x + 55y + 25(1000 - x - y)$
 $= 25x + 30y + 25000$

B1

1

(ii) Vertex check or cost lines $\Rightarrow$

M1

Minimised at C(500,0) with

A1

minimum cost = 37500

A1

Send 500 by road, 0 by rail, 500 by sea

B1

4

(iii) New cost $50x + Ry + 25(1000 - x - y)$

M1

$$= 25x + (R - 25)y + 25000$$

A1

The only vertex with $y \geq 500$ is A.

B1

Therefore comparing the gradients of the cost line and AB gives

$$\frac{25}{(R - 25)} > 3 \text{ or } R < 33\frac{1}{3}$$

So $R = 33$

A1

5

[18]

Q47.

(a) $x + y + z \leq 11$

B1

$$3x + y + 2z \leq 22$$

(or 4 100)

B1

2

(b) $P = 4x + 3y + z$

B1

1

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(c) $4x + 3y + z \rightarrow 4x + 3y + 1$

$$x + y + z \leq 11 \rightarrow x + y + 1 \leq 11$$

for both

B1

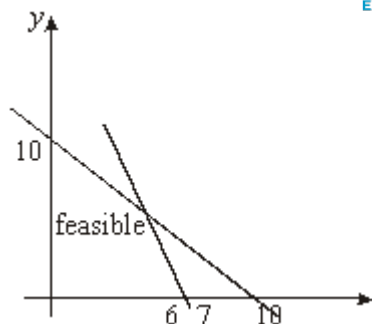
$$3x + y + 2z \leq 22 \rightarrow 3x + y \leq 20$$

for the new one

M1 A1

3

(d)



axes and general
one line
second line
region (ft)

M A1 A1 A1ft

Calculation of three vertices.

Or lines $4x + 3y = C$

M1

$4x + 3y + 1$ maximised at * ; (5,5)

A1

So maximum profit of £36

B1

by including 5 tins of chicken and 5 pots of caviar.

B1

8

(e) Trying $z = 0$ gives

M1

$$x + y \leq 11, 3x + y \leq 22$$

A1

Profit of $4x + 3y$ maximised at

M1

$$x = 5, y = 6 \Rightarrow P = 38$$

A1

4

[18]

Q48.

(a) (i) Flour $\Rightarrow 500x + 300y \leq 4500$

B1

Butter $\Rightarrow 50x + 150y \leq 1500$

M1

$\Rightarrow x + 3y \leq 30$

A1

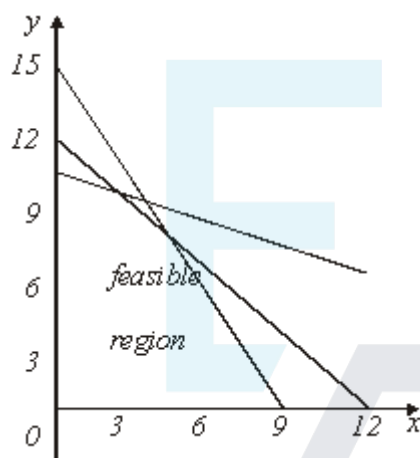
3

(ii) $x + y \leq 12$

B1

1

(b)



1 for each line and 1 for the feasible region

B1/B1/B1/B1

4

(c) Vertices: ©2025 Exam Papers Practice. All rights reserved.

$(0,0), (0,10), (3,9), (4.5,7.5), (9,0)$

(or by using lines and/or gradients)

M1

$P = 30x + 60y$ is maximised at $(3,9)$

ft first A1 A1 on sensible regions

A1 ft/A1 ft

So make 3 cottage and 9 farmhouse

A1

4

(d) $P = 90x + 60y$ maximised at $(4.5,7.5)$

M1/A1

Trying nearby integer points gives a maximum of 840p at (4,8) (and at (6,5))

M1

A1

4

[16]

Q49.

(a) $x + 2y \geq 60$

simplification not necessary

M1

$2x + y \geq 90$

any one

A1

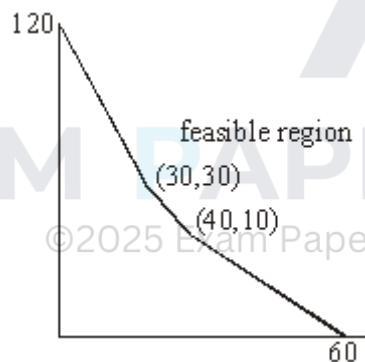
$3x + y \geq 120$

other two

A1

3

(b)



one line

other two lines

region

M1 A1 A1 A1

4

(c) x and y lines drawn

*or calculation at vertices,
but full method needed*

M1 A1

Minimised at (40,10), giving minimum

B1

number = 50

B1

4

- (d) From the graphs, the $x + 2y$ line must move up to meet $2x + y = 90$ where $x + y = 55$; i.e at (35,20) – they buy 35 Xtra and 20 Ybuild

any sensible method

M1

A1 A1

3

[14]

Q50.

- (a) $2 \leq x \leq y$

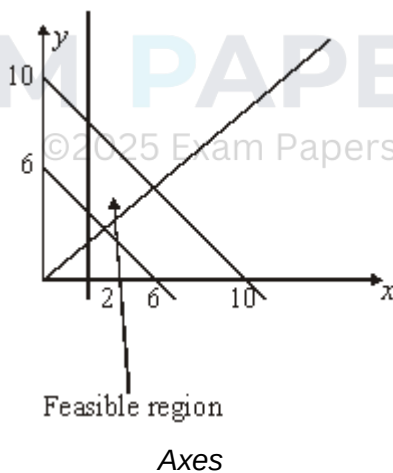
M1

$$6 \leq x + y \leq 10$$

A1 A1

3

- (b)



M1

4 lines @ A1 each, followed through from sensible answers to (a)

A1 A1 A1 A1

Region

A1

(c) $C = 2x + 4y$

B1

Either use of a cost line

or calculation of C at vertices:

M1

(3,3) £18 (2,4) £20

(2,8) £36 (5,5) £30

A1

Minimum cost £18

Maximum cost £36

Both correct

B1

4

(d) Integers 18, 19, ... , 36

M1

But must be even, so possible costs are
18, 20, 22, 24, 26, 28, 30, 32, 34 & 36

A1

2

EXAM PAPERS PRACTICE [15]

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Q51.

(a) $x \cdot \frac{1}{2} \text{ kg} + y \cdot 6 \text{ kg sugar} \leq 45 \text{ kg}$

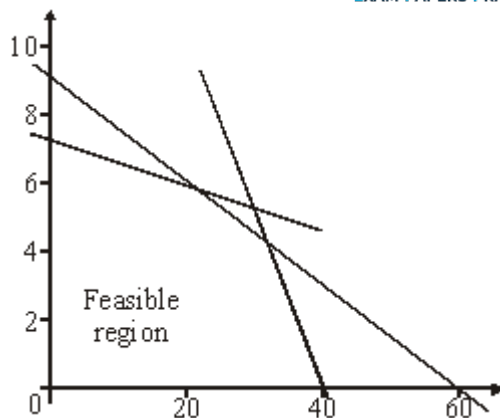
B1

$x + 7y \leq 60, x + 2y \leq 40$

B1 B1

3

(b)



One mark for each line
Region marked

B1 B1 B1 B1

4

- (c) Feasible region has vertices (0,7.5), (18,6)
(32,4), (40,0) and (0,0)

Or use of sliding line $J = 4x + 10y$

M1

A1

Jars needed = $2x + 10y$

M1

Maximum of 104 (at (32,4))

A1

4

- (d) Equal numbers of jars $\Rightarrow 2x = 10y$

M1

$$\Rightarrow y = 0.2x$$

M1

That line leaves feasible region at (25,5)
so he can make a maximum of 50 jars
of each.

M1

A1

4

[15]

Q52.

- (a) x = number of sets of pans

y = number of sets of crockery

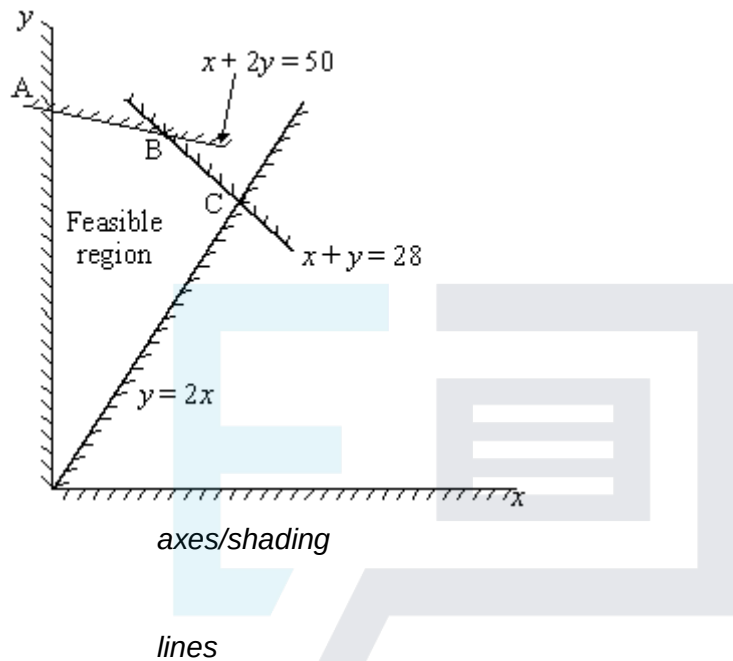
M1

$$y \geq 2x, \quad x + y \leq 28, \quad x + 2y \leq 50$$

A1A1A1

4

(b)



M1A1

A1A1A1

5

(c) $\{P = 2x + 3y$

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B1M1

{Profit line/vertex method $\Rightarrow$ B

A1

intersection of $x + 2y = 50$

$$x + y = 28$$

$$x = 6, y = 22$$

A1

Profit max = £78

B1

5

(d) New profit line = $P = 4x + 3y$

B1

$$\Rightarrow C: x + y = 28, y = 2x$$

M1

$$x = 9\frac{1}{3}, y = 18\frac{2}{3}$$

A1

Best integer solution 9, 19

M1A1

5

[19]

Q53.

(a) $\text{Max } N = 8x + 6y$

B1

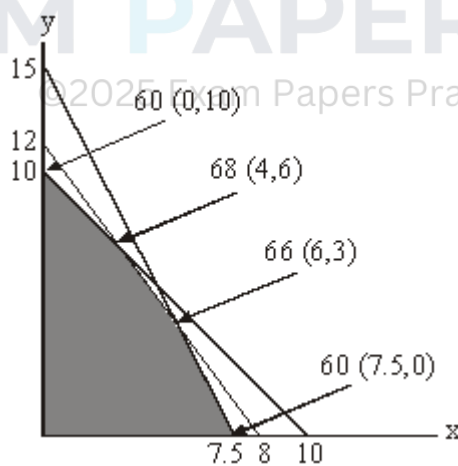
st $100x + 100y \leq 1000$
 $300x + 200y \leq 2400$
 (–1 each error)

B2

3

$$2x + y \leq 15$$

(b)



axes clear

M1

constraint lines
 (–1 each error/omission)

A2

Shading

B1

4 lots of the pancake recipe
valid method for choosing a correct vertex

M1

6 lots of the Yorkshire pudding recipe
68 items

Cao

A1

6

(c) No change - only using 14 eggs
cao

B1

1

[10]

Q54.

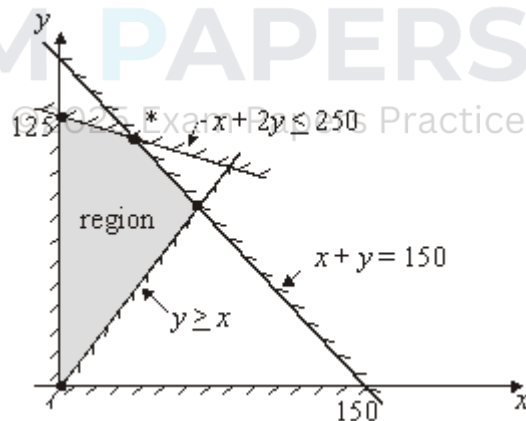
(a) $\geq 50\%$ German $\leftrightarrow$ German $\geq$ French

B1

$$x + y \leq 150 ; x + 2y \leq 250$$

B1B1

(b)



M1

shading etc

A1

Lines

A1A1A1

(c) $P = 1000x + 1500y$

can be implied

B1

max at *

profit line or vertex check

M1A1

Coordinates: $x + y = 150$ } $x = 50, y = 100$
 $x + 2y = 250$ }

B1

Max income = £200000

B1

- (d) Line $y = x$ replaced by $y = kx$ will not affect answer until * is below line

M1

At * $x = 50, y = 100, 66\frac{2}{3}\%$ are German

A1

$\therefore$ Can increase to $66\frac{2}{3}\%$

allow marks for cao with no working

A1

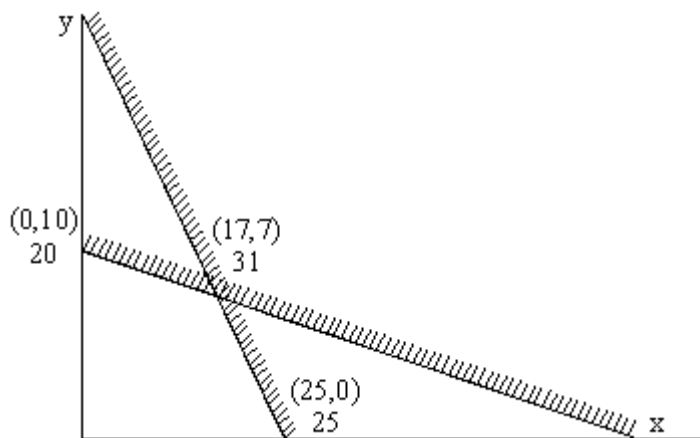
16

EXAM PAPERS PRACTICE [16]

Q55.

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(a)



scales and labels

B1

Lines

B1 B1

Shading

B1

Solution

B1

(b) $P - x - 2y = 0$ $P - \frac{11}{17}x + \frac{2}{17}s_1 = 20$
 Sca

M1

pivoting on $17y$

A1

$3x + 17y + s_1 = 170$ $\frac{3}{17}x + y + \frac{1}{17}s_1 = 10$
 (-1 each error)

A2

(0,10)

B1

$7x + 8y + s_2 = 175$ $\frac{95}{17}x - \frac{8}{17}s_1 + s_2 = 95$

or $P - \frac{8}{7}y + \frac{1}{7}s_2 = 25$
 sca

©2025 Exam Papers Practice. All rights reserved. M1

pivoting $7x$

A1

$\frac{95}{7}y + s_1 - \frac{3}{7}s_2 = 95$

(-1 each error)

A2

(25,0)

B1

(c) both = 0

B1

[11]

Q56.

(a) $\text{Max } 80c + 350t$

B1

st $20c + 100t \leq 10000$

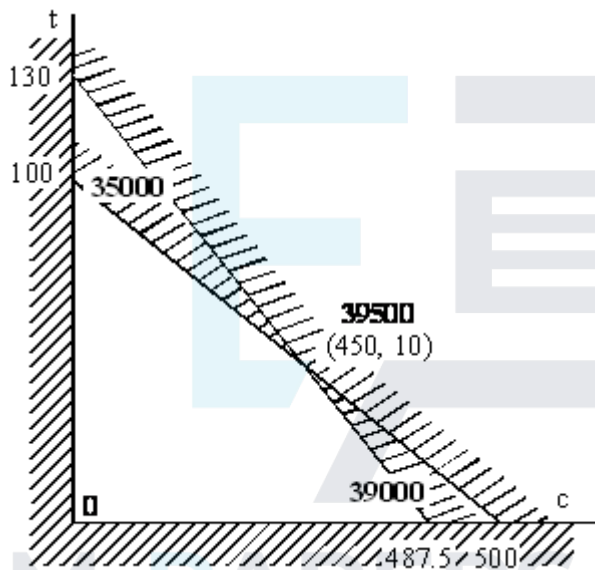
B1

$4c + 15t \leq 1950$

B1

$c, t \geq 0$

(b)



axes labelled and scaled

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B1

finance line

B1

hours line

B1

shading

B1

*method for optimising
(as indicated or by income lines)*

M1

450 chairs and 10 tables

A1

(c) unused finance and unused hours both zero

B1 both

B1

- (d) too many chairs per table
(or reasonable alternatives, eg. Lack of availability of tables)

B1

[12]



EXAM PAPERS PRACTICE

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Examiner reports

Q1.

This was a question which allowed students to demonstrate their understanding and application of mathematics in an unfamiliar situation, which is a key feature of the new specification. The majority of students scored 2/2 in part (a)(i). The most common error was to select an incorrect region as the feasible region. Some students did not draw the line $y = x$. Less than half of the students scored 2/2 in (a)(ii), with the most common mistakes being selecting the wrong vertex. The best solutions included an objective line which made the selection of the optimal vertex easier.

Nearly three quarters of students were able to explain the term 'connected' and relate it to the inequalities in (b)(i). However, less than half of students were able to correctly explain how the graph being simple related to the inequalities. Errors that were made included no direct reference to the degree of the vertices or the particular property of the graph that was being used.

Only 6% of students scored both marks in (b)(iii), with many students erroneously restating what they had written in (b)(ii). There were a significant number of no responses on this part. 10% of students scored the mark in part (b)(iv), with common errors being the adding together of two inequalities or the stating of a previous inequality.

Only a third of students scored any marks in (c)(i), with the vast majority not using the sum of the degrees being equal to twice the number of edges in the graph. Many answers did not write the values of x and y as pairs, and some solutions included pairs of x and y values that did not satisfy all of the inequalities.

More than half of students were able to score at least one mark in (c)(ii), with many students scoring 2/2 having not scored full marks on (c)(i). The better solutions included writing down the degree of each vertex next to the respective vertex. The loss of marks was most commonly due to having too few or too many edges on the graph.

Q3.

Part (a) was very well answered, and this showed a significant improvement on previous papers. A few students contributed to their own downfall by the use of very faint, spaced out shading – this often made it appear that the small triangle below the line $y = 20$ was to be included in the feasible region. In part (b), marks were lost for not stating the x and y values, and in (ii) for thinking that integer values were needed despite this constraint not having been specified. Students also showed a marked reluctance to use fractions.

Q4.

The response to part (a) was very good and most students earned full marks. Answers to (b)(i) were much improved from previous years. The majority of students made the transition from one inequality to the other very clear, including the actual substitutions required. In (b)(ii) graph drawing continued to be of a good standard with very few examples of slipshod and inaccurate execution. The objective line, as usual, caused the most problems. A minority of students produced a line through the origin as their apparent objective line. Unfortunately a number of these then shaded on one side or the other of this line thus rendering their feasible region incorrect. Questions (b)(iii) and (c) proved to be by far the most difficult and some students did appear to be rushed in their working. Roughly half the students solved the problems correctly but most of these failed to earn full marks either by not specifying the unit of currency or by not describing

precisely and accurately the numbers and types of each bouquet required. The question is set in a context and answers must reflect this to gain full credit.

Q5.

Part (a) was well-answered. Almost all candidates had the five correct lines. A minority failed to draw each of them to an acceptable accuracy and another small group did not label the feasible region. Part (b) was not answered quite as well, with written and drawn evidence of at least one objective line. A few failed to realise that the values of x and y should be integral.

Q6.

In part (a) most students successfully wrote down enough for the first 2 marks but again using percentages defeated a surprising number of students.

Even for those students who managed the first part, the substitution of $z = 0.5x$ in part (b) proved too difficult for all but the strongest students.

Graphs are still not always accurately drawn, and many were unable to draw the graph of $y = x$ due to the different scales used on the axes. Also, a surprising number of students omitted to draw the lines $x = 100$ and $y = 200$.

The last two parts proved too much for all but the most able. Only the strongest students were able to find the objective function and identify the correct numbers of each type of pillow. Some also insisted on not writing this in context and so were not awarded the marks.

Q7.

The ability to use algebra that was required in parts (a) and (b)(i) was much improved on past performance. The commonest errors were to forget to simplify in part (a) or to fail in that part to realise that one inequality was 'the other way round'. Presentation for the first required inequality in part (b)(i) was much improved, but a minority failed to identify clearly the source of the remaining two.

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Graph plotting in part (b)(ii) was much improved, with the exception of misreading the x -axis scale when plotting the intercept for the line $3x + 5y = 200$. However, to counterbalance this there was quite a lot of simply shoddy and inaccurate drawing. Not only did this lose marks here but it made mark-gathering in later parts more difficult.

Most candidates found the last two parts very difficult. Few realised that, in one way or another, the expression $x + 2y$ must be maximised. Far more thought that $x + y$ must be maximised. Many of those who were more or less on the right path went astray by halving their value of y for part (b)(iv).

Q8.

Parts of this question proved to be very accessible to most candidates. The great majority scored 3 or 4 marks on part (a), the lost mark almost always being because of the omission of the function to be minimised. Many candidates lost marks on the graph. The different scales caused problems, especially on the x -axis, with 11 being marked where 12 was clearly intended, being a common error. These problems were exacerbated by a surprising number of slipshod errors in actually drawing the lines.

Candidates should realise that examiners have to make judgements about accuracy. Some tolerance is allowed but many candidates failed to draw inside this and consequently lost marks. Regardless of success at the graphs, most candidates managed to find the correct values required for x and y , whereupon a few promptly lost a mark by failing to refer back to the context. A smaller number lost another mark by failing to specify the currency unit.

Q9.

This style of question, as before, differentiated between types of candidates, as some sub-GCSE level graph drawing appeared. Generally, though, candidates were more careful with their drawing, and there were far fewer instances where inaccuracies could justifiably be penalised.

Part (b) was poorly answered, right across the ability range. Many simply offered no solution at all. For those who did offer answers, it appears that the phrase 'use your diagram' caused candidates to concentrate too much on sliding rulers across their graph and on reading the coordinates of their points that they completely failed to notice that the points that they were homing in on did in fact have integral co-ordinates. There was also a widespread failure to give both coordinates and maximum values or to indicate which went with which.

Q10.

Most candidates failed to score 10 or more marks – a very disappointing performance from candidates.

Part (a) Most candidates scored the first 3 marks although $x + y + z \geq 300$ caused more problems than it should. The fourth, involving percentages was beyond most. A significant minority clearly earned the mark for this by working back from the required answer in the next part.

Part (b)(i) Most candidates failed to satisfy the requirement for the M mark and most of those who did failed to earn the second mark – often this was because of previous failure with the percentage.

Part (b)(ii) Graph drawing still seems to be improving, with most candidates scoring at least 5 marks but seldom more. An objective line was usually absent and if present was usually a line representing income, not profit. Failure to label the feasible region was still quite common and careless line-drawing was not uncommon.

Part (b)(iii) Zero was by far the most common mark. As already mentioned, working with income rather than profit was a major error and those who, by hook or by crook, identified (320, 50) as the relevant point usually failed to state the number of each type of ball to be manufactured or specified 25 each of medium and fast rather than 50 of each.

Q11.

Overall this question discriminated well.

In part (a), most candidates scored at least 4 marks but the failure to address the profit was the primary cause for not scoring full marks. There were a few errors with the inequalities.

There was an encouraging response to part (b)(i). Most candidates appreciated the need to make their method absolutely clear (the answers were given) and were fully capable of this. In part (b)(ii), the standard of graph drawing in questions like this is still improving. However, it is still surprising to see the number of complete failures, from candidates, to draw correctly the graph of $2x + y = 300$ or even $y = x$. For the objective line a gradient of $-3/7$ was just about as common as the correct $-7/3$. A small number of candidates thought that the “direction of the objective line” required only a casual arrow in the general direction of the slide of the line once it had been drawn rather than an accurate drawing of a possible position for it.

In part (c), a significant number of those who got this far failed to work carefully enough with the objective line and identified the wrong point from the two obvious choices. Another failing was to not identify the solution explicitly and completely in the context of the question: numbers of all three animals and the profit.

Q12.

Many candidates demonstrated a considerable lack of algebraic ability in this question, whereas more able candidates coped well with much of parts (a), (b)(i) and (b)(ii). However, their almost total failure to cope with the rest of the question was a great disappointment. In part (a), the first three inequalities were found and simplified by almost all the more able candidates; the ‘average’ candidate usually found the inequalities but neglected to simplify them; but very few candidates at all got anywhere near the fourth inequality. The most common inequalities were often devoid of x , y and z , and some were sometimes randomly combined. A few had the correct idea, but had the signs the wrong way round. In part (b)(i), only a minority offered any answers, and of these, only some presented their work clearly enough to earn any marks. Despite this, part (b)(ii) was comparatively encouraging, apart from drawing of the line $y = x$: many thought that this was drawn simply by joining $(0, 0)$ to $(40, 60)$. Many candidates also failed to label their feasible region. The majority of candidates omitted parts (b)(iii) and (b)(iv). Those who did offer answers seemed to ignore the implications of $x = z$. Most of the final answers never involved z and virtually no candidates made any attempt to relate their answers to the context.

Q13.

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This question was not well done. In part (a), a large number of candidates could not draw the graph of $y = \frac{1}{2}x$ correctly. Many drew the graph of $y = 2x$ and even more a graph of gradient $-\frac{1}{2}$ that didn’t pass through the origin. Even those with the ‘correct’ graph lost marks through drawing it inaccurately.

In part (b), many candidates assumed that the point $(10, 20)$ must be involved, often for both parts, and some appeared to make no use of objective lines whatsoever. A significant minority, including many able candidates, were content to give only coordinates and failed to calculate the corresponding values of F .

Q14.

This question did differentiate to reveal the more mathematically able. The weakest candidates were unable to distinguish between A , B and C , and x , y , and z ; other candidates confused ‘rows’ and ‘columns’.

There was, however, a number of candidates who were able to obtain the first three inequalities and many of these obtained the fourth and were able to simplify it. However, the final inequality proved to be beyond the reach of many; its simplified answer was

obtained by just a few of the very best candidates.

Q15.

This question discriminated well.

Part (a) was done quite well. Only the weakest candidates failed to use the information correctly although a considerable number failed to state the function to be minimised.

Part (b) discriminated well. It continues to be the case that many candidates who appear to have no problem in writing down the inequalities and understand they must draw the lines representing these simply cannot draw the requisite lines. Some even drew the wrong lines parallel to the axes. In otherwise fairly successful work the feasible region was not always clearly indicated.

Part (c) caused major problems and very few candidates scored all 4 marks. The majority simply did not state clearly how they were trying to solve the problem, ie which point(s) they were looking at. This was important as the initially required point does not have integral coordinates. From those who did realise that an intersection point was needed, there was some surprisingly incompetent simultaneous equation work. Most of those who did find the correct coordinates then proceeded to use them and not look for qualifying points with integral coefficients.

Q16.

All candidates were able to score on this question although fully correct solutions were rare. Parts (a) and (b) were well answered. In part (c), drawing the lines $y = 2x$ and $y = 3x$ proved to be beyond a significant number of candidates. Some candidates who correctly drew all the lines failed to clearly indicate the feasible region. In part (d), many candidates drew 'profit' lines. A surprising number of candidates failed to find the exact value of the intersection. Although the graph would indicate the maximum point, candidates should solve the simultaneous equations to find the exact values of intersections. Candidates must check that their answer is sensible.

Q17.

In part (a), the inequalities $x \geq 50$, $y \geq 50$ and $x + y \geq 140$ were often not clearly and precisely enough explained: some thought $x \geq 50 \Rightarrow x \geq 50$ would suffice! Others perhaps thought they were too obvious to bother about.

Responses to part (b)(i) were quite variable. Many candidates, including those who were otherwise very able, drew the lines very poorly; some were completely defeated by them. On the other hand, there were a number of candidates who did poorly on most other questions – including the rest of this one – but here produced complete, neat and accurate graphs. The scale caused problems, particularly dealing with 140. About half the candidates failed to draw an objective line, and for those who drew the objective line many had an incorrect gradient.

Part (b)(iii) differentiated quite well. Most candidates who answered this appreciated that a vertex should be considered but often did not consider enough or had no objective line to help. A few did not realise the need for integral coordinates.

Teachers may wish to work with students on:

- drawing straight lines which do not cross the axes,
- drawing objective lines,
- the use of integer solutions where appropriate to a model.

Q18.

This question proved to be a good discriminator. In part (a) candidates realised that they had to consider extreme points on the feasible region, but many failed to find the correct vertices.

In part (a)(iii) only the best scripts correctly identified (75, 10) as the minimum point on the feasible region. It was pleasing to note the number of candidates who scored full marks in part (b), although the inequality $2x + y < 160$ was the most challenging for candidates who failed to score full marks.

Q19.

In part (a) candidates realised that they had set up the inequalities and this part was well answered, but drawing the two diagonal inequalities proved too difficult for many. Centres must ensure that candidates do not always rely on equating x and then y to zero to draw the graphs, as the scale may prevent this. Many students omitted drawing an objective line even though this is clearly mentioned in the specification. Candidates could still consider extreme points on the feasible region for the final two parts of the question, but many failed to find the correct vertices.

Q20.

All candidates were able to score on this question. Parts (a) and (b) were well answered. Many candidates were unable to handle the difference in the scales on the axes and a large number of candidates failed to produce a correct objective line. There were very few candidates that were able to answer part (e) correctly; this proved to be a good discriminator for the grade A candidates. The popular wrong answer for part (e)(ii) was 160.

Q21.

All candidates were able to score on this question. Only the best candidates were able to produce convincing arguments in part (a). Parts (b) and (c) were well answered with diagrams being clearly drawn although a significant number failed to draw the objective line.

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Q22.

All candidates were able to score on this question, though only the best candidates were able to produce convincing arguments in part (a). Part (b) was answered reasonably well with diagrams being clearly drawn. Many candidates failed to draw the line $y = \frac{2}{3}x$ and this led to errors in part (c). Many candidates also failed to produce a correct objective line. Part (d) was usually well answered by candidates who persevered with the question.

Q23.

All candidates were able to score on this question. Only the best candidates were able to produce convincing arguments in part (a). Parts (b) and (c) were well answered with diagrams being clearly drawn. However there was a significant number of candidates who could not draw the line $2y = x$. There were very few candidates that were able to answer part (e) correctly; this proved to be a good discriminator for the grade A candidates. Candidates needed to realise that although an integer solution was required, this was only for the seventh hour of the day.

Q24.

This question was reasonably well attempted by the majority of candidates. Many scored full marks in part (a), but were unable to explain how their inequalities could be converted to the forms required in part (b)(i). Many candidates scored full marks in part (b)(ii), with the feasible region clearly shown, but weaker candidates often misinterpreted the scales given on the insert.

Q25.

Most candidates were able to earn marks on this question. Only the best candidates produced convincing arguments in part (a).

Parts (b) and (c) were well answered with diagrams clearly drawn. Many candidates did not draw a correct objective line.

There were very few scripts in which part (d) was answered correctly; this part proved to be a good discriminator for the grade A candidates.

Q26.

This question was answered better than on previous occasions with a large number of candidates scoring full marks. A number of candidates reversed the inequalities that only included terms in x and y . (For example: $x \leq 2y$ or $y \leq 3x$). In part (b) candidates realised that they had to consider extreme points on the feasible region, but many failed to find the correct vertex at (8, 16).

Q27.

This question was well answered by the majority of candidates. There were very few candidates who used A and B instead of x and y .

In part (a) many candidates omitted the objective function. Many candidates scored full marks in part (b), where the feasible region was usually clearly shown, even when incorrect. Part (c)(i) was well answered, but there were very few correct answers to part (c)(ii), where the most common wrong answers were 2, 1 and 40.

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Q28.

Candidates were able to make a good attempt at this question. Candidates made mistakes in the inequalities in part (a), but were able to recover and draw a correct diagram.

The incorrect answer of $x + y < 12$ was common, yet the correct graph of $3x + y < 12$ was often then drawn

Part (c)(i) was well answered, but there were fewer correct answers to part (c)(ii).

Q29.

Candidates were able to answer parts (a) and (b)(i) successfully, although all but the best candidates failed to find the minimum value of $-x + 2y$ in the final part. As the graph was given, it would have been acceptable for candidates to substitute the extreme co-ordinates into the function to find the minimum.

Q30.

This was the first time a linear programming question had been set in this format and the responses were very good. Part (a)(i) was well answered but a number of candidates found the maximum value in part (a)(ii) as opposed to the minimum value that was requested. There were good attempts at finding the inequalities in part (b). The one proving to be most difficult was

$2x + 5y \leq 600$ which was frequently incorrectly stated as $x + 2.5 \leq 120$.

Q31.

This was a routine linear programming problem but there were very few fully correct solutions seen. A significant number of candidates used A and B to form their incorrect inequalities. For the candidates who had the correct inequalities, part (b) was a good source of 6 marks. The majority of candidates could find the minimum daily income but the maximum defeated all but the most able candidates. It was disappointing to see that many scripts stated the company should make 4.6 and 10.2 televisions. To score full marks, candidates had to establish that the best integer solution was required and test integer solutions around the point (4.6, 10.2).

Q32.

This was usually well answered. Candidates used the pre-scaled graph paper and drew clear diagrams. Marks were lost in part (b) by candidates drawing an incorrect objective line in part (a). Candidates could still have scored full marks in part (b) if they had considered extreme points.

Q33.

The general presentation of graph work in this question was pleasing, although some candidates failed to identify the feasible region clearly and a significant number failed to draw an objective line. Some candidates answered part (c) by considering all points of intersection and evaluating the income at each of these points. This method, although more time consuming, was acceptable as long as the candidate clearly indicated their minimum and maximum.

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Q34.

The majority of candidates produced inequalities in three variables and part (a) was answered quite well – although many showed a lack of basic knowledge of percentages. There were weaknesses in the simplification of inequalities so that there were only integer coefficients. Many candidates were able to substitute $y = 6$ and made some progress in part (b) but there were very few completely correct solutions.

Q35.

There was an improvement compared to the previous paper in the presentation of work, with many excellent graphs and the feasible region clearly labelled. In part (a) the explanations were very imprecise, especially as to whether a statement should or should not be a strict inequality. Part (c) required candidates to list all values, not some values which produced the minimum profit. Part (d) was found to be very demanding but it was very pleasing to see some candidates producing perfect algebraic solutions.

Q36.

It was pleasing to note the presentation of the graphical work done by the vast majority of

candidates, including the labelling of an objective line. In part (d) only the better candidates realised that the maximum number of items required integer answers - many candidates were quite happy to record that 55.7 jumpers were made. When considering integer solutions, candidates should be advised to show that they have considered points around the optimum solution before discounting points not in the feasible region.

Q37.

Candidates were able to make a good attempt to this question, although some of the later parts were answered less well. In part (a) a number of candidates wrote down that the first two inequalities were trivial without relating this fact to a real life situation. Part (b) required a graph to be drawn. Advanced level candidates should appreciate the need for a suitable scale, but some candidates chose a scale so small that correct solutions to the later parts of the question became impossible to achieve. The majority of candidates 'shaded out' to produce a feasible region but they should have clearly indicated which was the feasible region on their diagram. Part (b)(iv) proved to be challenging for the candidates producing an answer that was outside the feasible region.

Q38.

It was pleasing to note the improved presentation of the graphical work of candidates, the only criticism being of the lack of labelling of an objective line. In part (b)(iii) only the better candidates realised that the maximum number of items required the maximum value of $x + 2y$. Part (b)(v) proved to be a good discriminating request.

Q39.

Candidates were able to make a good attempt to this question, although some of the later parts were answered less well. In part (a)(ii) a number of candidates wrote down two non-negative inequalities, perhaps because this will be true in all real life situations. Part (b) required a graph to be drawn. A level candidates should appreciate the need for a suitable scale, but some candidates chose a scale so small that correct solutions to the later parts of the question became impossible to achieve. The majority of candidates 'shaded out' to produce a feasible region but they must clearly indicate which is the feasible region on their diagram.

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Q40.

This question was answered well. In part (a), candidates had to justify the results and not merely copy the inequalities from the question. In part (b), candidates chose suitable scales but there were still some candidates who did not indicate their feasible region despite this being a specific request in the question.

Q43.

Candidates who had covered linear programming thoroughly found this question relatively straightforward, with a minority taking the wrong region for this minimising problem. However, there are still some candidates who seem under-prepared for this topic and who barely make a start on it.

Q44.

The unusual style of the first few parts of this question avoided the need to draw graphs and this benefited the candidates, most of whom answered parts (a), (b) and (c) correctly. The most common error in those parts was to locate the optimum point (2, 14) but then fail to draw the practical conclusion. The more discriminating part (d) did require candidates to

think about the effect of an additional line $4y = x$, but most found this rather hard.

Q46.

It was unfortunate that many candidates failed to realise that the number of cases sent by sea is merely $1000 - x - y$. Failing to realise that meant that candidates lost the marks from part (a), but the question was designed so that they could continue anyway, with the necessary linear equations clearly stated in part (a). However, surprisingly many candidates then failed to draw the three straight lines needed in part (b). Consequently only a minority of candidates succeeded in part (c), and the discriminating final part. Once again, after some improvement in recent papers, the general performance in linear programming was disappointing.

Q47.

The performances on this question were very mixed. For those candidates who were familiar with linear programming, most of this question caused very little problem, with just the last part proving to be discriminatory. Another example of candidates failing to read the question meant that generally an easy mark was lost in part (c) for failing to justify the expressions $P = 4x + 3y + 1$ and $x + y \leq 10$.

Q49.

This was perhaps the most disappointing question because on previous specifications this topic was always well done. The candidates' performances showed a general lack of preparation. It was hoped that giving the axes on an insert would help, but even so many candidates labelled the wrong region as being feasible and very few gave a systematic method for minimising $x + y$.

Q50.

This linear programming question was very well done on the whole. Even the last part (where the answer was simply all the even numbers between the minimum and the maximum) was answered well, although sometimes after a lot of wasteful effort. When a part of a question is only worth 2 marks candidates should realise that not a lot of work is expected. In general, in linear programming questions, there is one poor aspect of the notation which many candidates use: when referring to the point (2, 8), say, rather than using coordinates or calling it $x = 2$, $y = 8$, they often refer to it as $2x + 8y$, which of course is not the same thing at all. This was not penalised but should definitely be avoided!

Q52.

This question produced the most disappointing responses. In the past, candidates have always scored highly on linear programming. But this time, however, most of the attempts were very sketchy and showed a real lack of practice on this topic.

Q54.

This was a fairly standard question tackled reasonably well by most candidates, but it was surprising how many of them only included two out of the three lines in their graphical solutions. Again, the last part succeeded in discriminating between the best candidates.

Q55.

The graphical solution in part (a) of this question was very well done, with many

candidates obtaining full marks. Far fewer candidates could apply an iteration of the simplex algorithm in part (b) correctly, although some excellent solutions were seen. Few candidates could identify the point on the graph that corresponded to the result of the iteration. More candidates knew that the slack variables would both have value zero at the end.

Q56.

Most candidates scored well on this question. Not all formulations were complete - "maximise" was often omitted. There was often no justification given for choosing 450 chairs and 10 tables. In part (c) good explanations of slack variables were rarely seen. In part (d) most candidates saw the obvious practical problem, but quite a few did not.



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